# PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

**(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)**

# SYLLABI

*FOR*

**B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)**
**Discipline Specific Core Papers**

**FIRST YEAR**
**(SEMESTER SYSTEM)**
**EXAMINATIONS, 2025-26**

**(SEMESTER : FIRST AND SECOND)**

*i.e*

***First Semester:*** ***November/December, 2025***
***Second Semester:*** ***April/May, 2026***

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# Subject combinations offered in the Faculty of Science under NEP

| Discipline Specific Courses (DSC) | | | |
|---|---|---|---|
| **1** | **2** | **3** | **4** |
| Physics | Chemistry | Mathematics | Bio-informatics |
| Botany | Computer Science and Applications | Zoology | Biotechnology |
| | Statistics | | Industrial Chemistry |
| | | | Agriculture |
| | | | Fashion Designing |
| | | | Bio-chemistry |
| | | | Microbiology |
| | | | Microbial Food Technology |

**Note: Candidate will choose three subjects by selecting one course from each of the columns 1 to 4.**

# Table-2

## NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY B.SC, B.SC HONOURS AND B.SCHONOURS WITH RESEARCH

| Semester | Subject- A (4 Cr) | Subject-B (4 Cr) | Subject- C (4 Cr) | MDC /IDC (3 Cr) | AEC (2 Cr) | VAC (2 Cr) | SEC (3 Cr) | Internship/ Community Work (4 Cr) | Pbi/ HCP (2Cr) | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| I | MAJ A-1 | MAJ B-1 | MIN C-1 | MDC-1 | Eng-1 | VAC-1 | SEC-1 | | Pbi-1/ HCP-1 | 24 |
| II | MAJ A-2 | MAJ B-2 | MIN C-2 | MDC-2 | Eng-2 | VAC-2 | SEC-2 | | Pbi-2/ HCP-2 | 24 |
| **The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summerinternship/community work.** | | | | | | | | | | |
| **The candidate will have to declare one of the three subjects as Minor in the 3rd semester. The other 02 subjects will remain his/her Major subjects** | | | | | | | | | | |
| III | MAJ A-3 | MAJ B-3 | MIN C-3<br>MIN C-4 | MDC-3 | Eng / MIL | | SEC-3 | | | 24 |
| IV | MAJ A-4<br>MAJ A-5 | MAJ B-4<br>MAJ B-5 | MIN C-5 | | Eng/ MIL | VAC-3 | | | | 24 |
| **The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship** | | | | | | | | | | |
| V | MAJ A-6<br>MAJ A-7 | MAJ B-6<br>MAJ B-7 | MIN C-6 | | | | | Internship/ Community Work | | 24 |
| VI | MAJ A-8<br>MAJ A-9 | MAJ B-8<br>MAJ B-9 | MIN C-7<br>MIN C-8 | | | | | | | 24 |
| **The student successfully securing 144 credits will be awarded BSc degree** | | | | | | | | | | |
| | 36 | 36 | 32 | 09 | 08 | 06 | 09 | 04 | 04 | 144 |
| VII | MAJ A-10<br>MAJ A-11 | MAJ B-10<br>MAJ B-11 | MIN C-9<br>MIN C-10 | | | | | | | 24 |
| VIII | MAJ A-12 OR RM<br><br>MAJ A – 13<br>MAJ A – 14<br>Or<br>Dissertation of 08 credits | MAJ B-12 OR RM<br><br>MAJ B – 13<br>MAJ B – 14<br>Or<br>Dissertation of 08 credits | | | | | | | | 24 |
| | 56 | 56 | 40 | 09 | 08 | 06 | 09 | 04 | 04 | 192 |

Notes:-

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. Thestudent choosing Single Major stream is required to give his/her Major and Minor at end of the second semester.
3. The student choosing Multi-Disciplinary stream will continue with the three subjects and complete his/her three year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3 year degree) shall be eligible for 4 YEAR DEGREE B.SC HONOURSwith research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two year securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of08 credits in one of the two Major subjects.

# Foreword

This document provides the syllabi for the first two semesters of various science subjects in the category of Discipline Specific Courses (DSC) for the first year students enrolled in four year degree programme in Science stream (except B.Sc. Biochemistery, B.Sc. Bioinformatics, B.Sc. Biotechnology, B.Sc Microbial and Food Technology and BCA) under NEP. The Syllabi of other categories i.e. MDC, AEC, VAC, SEC, Copulsory Punjabi and HCP have been given separately.

The University has approved two distinct templates leading to two types of degrees: Subject-Specific and Discipline-Specific. This document is alligned with the Template-2 which is Multidisciplinary Stream.

# Agriculture

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY**

**B.A./B.Sc., B.A./B.Sc. HONOURS and B.A./B.Sc. HONOURS WITH RESEARCH IN AGRICULTURE**

**(Table 2*)**

| Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | Total Credits |
|---|---|---|---|---|---|
| I | L-100 | AGR-DSC-1-Maj/Min-101 | Elements of Agronomy and Agriculture | 3+1=4 (x3) | 24 |
| | | AGR-MDC-1-101 | Agricultural Botany & Crop Physiology | 2+1=3 | |
| | | AGR-VAC-1-101 | - | 2 | |
| | | AGR-SEC-1-101 | Nursery Production | 2+1=3 | |
| | | AEC-1-101 | Eng-1 | 2 | |
| | | PUN-COMPL-1-101/HCP-COMPL-1-101 | Pbi-1/HCP-1 | 2 | |
| II | L-100 | AGR-DSC-2-Maj/Min-201 | Basics of Agricultural Economics | 3+1=4 (x3) | 24 |
| | | AGR-MDC-2-201 | Agricultural Microbiology | 2+1=3 | |
| | | AGR-VAC-2-201 | - | 2 | |
| | | AGR-SEC-2-201 | Biofertilizers | 2+1=3 | |
| | | AEC-2-201 | Eng-2 | 2 | |
| | | PUN-COMPL-2-201/HCP-COMPL-2-201 | Pbi-2/HCP-2 | 2 | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II | | | | | |
| The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. | | | | | |

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3$^{rd}$ semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## B.A./B.Sc.(General) Agriculture (Major/Minor)

**Course Title: ELEMENTS OF AGRONOMY AND AGRICULTURE** **Course Code: (AGR-DSC-1 Maj/Min-101)**
**Semester: First**

**Learning Outcome:** To give student elementary information on basic agronomic principles and practices.

**Max. Marks: 100** **Course Credits** = 3+1=4

| | | |
|---|---|---|
| 1. Theory: 75 | Written Exam: 67 | Internal Assessment: 8 |
| 2. Practical: 25 | Written Exam: 22 | Internal Assessment: 3 |

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

**Instructions for Paper Setters:**
Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

### UNIT – I 12 Hours

Evolution of agriculture, ancient agriculture, history of Indian Agriculture, Principals of agronomy, classification of crops.
Crop rotation: principles, limitation, advantages; rotational intensity, cropping scheme, cropping intensity.
Cropping system: intercropping, mixed cropping, multiple cropping and relay cropping.

### UNIT – II 11 Hours

Agronomy as a science and its relationships with other sciences.
Germination, maturity, harvesting and storage of crop plants.
Physical &biological properties of soil, soil organic matter.
Tillage principles, tillage practices for different soil types and crops.

### UNIT – III 11 Hours

Seed bed preparation.
Characteristics of good seed beds.
Methods of sowing and their suitability under different conditions.
Seedling practices in relation to types of seeds, time of sowing and soil moisture.

### UNIT – IV 11 Hours

Weed characteristics, dissemination and competition for growth factors and problems caused by them.
Common weeds, losses and benefits, crop weed associations
Common methods of weed control.
Green manuring – advantages, limitations, green manure crops.

## GUIDELINES FOR PRACTICAL EXAMINATION

**Time- 3 Hrs** **Max Marks: 25**
**Practical: 22** **Internal Assessment: 03**

1. Land measurement
2. Identification and practices in the use of farm hand tools, bullock and tractor driven implements.
3. Identification of crop seeds, crop plants and important weeds.
4. Knowledge of various fertilizers – collection, weeding, hoeing in various crops.
5. Collection of weeds
6. Use of spray pumps.
7. Calculation of herbicide doses for different crops.

**Suggested Readings:-**

1. Reddy S.R., *Principles of Agronomy.*Kalyani Publishers, Ludhiana, 2018.
2. Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
3. *Weeds of North India* I.C.A.R. Publications, New Delhi, 2008.
4. *Package of Practices for Rabi and Kharif crops,* P.A.U. Publications Ludhiana, Corresponding year.

## B.A./B.Sc.(General) Agriculture (Major/Minor)

**Course Title: BASICS OF AGRICULTURAL ECONOMICS**

**Course Code: (AGR-DSC-2 Maj/Min-201)**
**Semester: Second**

**Learning Outcome:** To give student elementary information on basic principles of economics related to agriculture.

| | | |
|---|---|---|
| **Max. Marks: 100** | | **Course Credits = 3+1=4** |
| **1. Theory: 75** | **Written Exam: 67** | **Internal Assessment: 8** |
| **2. Practical: 25** | **Written Exam: 22** | **Internal Assessment: 3** |

**Periods per week:** 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

**INSTRUCTIONS FOR PAPER SETTER AND EXAMINERS:**
Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

### UNIT – I — 12 Hours

Agricultural Economic: Meaning, scope and its relationship with other social and agriculturalsciences.
Recourse Management in Agriculture: land,labour and capital.
Agriculture and Economic Development: Introduction, role, share of agriculture ineconomic development; Role of agriculture in economic development of India and its comparison with other countries.
Factor affecting agricultural development.

### UNIT – II — 11 Hours

Risk and Uncertainty: Introductions, risk types and risk as choice indicator; different theories of risk and uncertainty; measures to minimize risk and uncertainty.
System of Farm Organization: Introductions, types, criteria for choice in India and suitability of alternate systems.
Farm Size and Practices: Holdings and farm.
Dynamics of Population: Population growth and its effect on agricultural growth rate.

### UNIT – III — 11 Hours

Agricultural marketing, efficient marketing system, classification of markets, problems of Indian agricultural marketing and measure to improve agricultural marketing.
Rural Indebtness: introduction, extent, nature and causes of rural indebtness, Government measure to reduce rural indebtness
Agricultural Finance.
Agricultural credit institutions: structure of agriculture credit system, multi-agency approach, regional rural banks, NABARD, ARDC

**UNIT – IV** **11 Hours**

Self Help groups, cooperative credit structure-three tier structure
Rural Poverty: Introduction, factor responsible for poverty, suitable measures to reduce poverty and poverty alleviation programme.
Cottage and small industries: Features and classification of cottage industries. Difference between cottage and small industries, role and problems of cottage and small industries
Different measure to promote SSI and role of Government, small industries development bank of India and its role, khadi and village industries (KVIC)

## GUIDELINES FOR PRACTICAL EXAMINATION

**Time: 3 Hrs** **Max Marks: 25**
**Practical: 22** **Internal Assessment: 03**

1. Calculations of interests.
2. Calculations of depreciation.
3. Farm Budgeting
4. Preparations of balance sheet and other final statements of financial importance for agriculture farm.
5. Study tours of Marketing Centers of local area.

**Suggested Readings:**

1. Lekhi, R.K. and Singh, J., Agricultural Economics, Kalyani Publishers, Ludhiana, 2016.
2. Black. J.D., *Introduction of Economics for Agriculture,* Fromount Pierre National Press, 1955.
3. **Nanavati, M.B. and J. J. Anjaria, The Indian Rural Problem. The Indian Society of Agricultural Economics, Bombay, 1944.**
    1. Memoria, C.B. and B.B., *Agricultural Problems in India,* KitabMahal, Allahabad, 2007.

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# BIO-CHEMISTRY

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)**
**and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)**

**B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BIOCHEMISTRY**

| Table 1* | | | | | | Table 2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** |
| I | L-100 | (BCH-DSC-1 Maj/Min-101) | Biomolecules I | 3+1=4 | 24 | I | L-100 | (BCH-DSC-1 Maj/Min-101) | Biomolecules I | 3+1=4 (x3) | 24 |
| | | (BCH-MDC-1-101) | Introduction to Biochemistry | 2+1=3 | | | | (BCH-MDC-1-101) | Introduction to Biochemistry | 2+1=3 | |
| | | (VAC-1-101) | | 2 | | | | (VAC-1-101) | | 2 | |
| | | *(BCH-SEC-1-101) | Biochemical Techniques-I | 1+2=3 | | | | *(BCH-SEC-1-101) | Biochemical Techniques-I | 1+2=3 | |
| | | (AEC-1-101) | English | 2 | | | | (AEC-1-101) | English | 2 | |
| | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | | | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (BCH-DSC-2 Maj/Min-201) | Biomolecules II | 3+1=4 | 24 | II | L-100 | (BCH-DSC-2 Maj/Min-201) | Biomolecules II | 3+1=4 (x3) | 24 |
| | | (BCH-MDC-2-201) | Fundamentals of Enzymology | 2+1=3 | | | | (BCH-MDC-2-201) | Fundamentals of Enzymology | 2+1=3 | |
| | | (VAC-2-201) | | 2 | | | | VAC-2-201) | | 2 | |
| | | *(BCH-SEC-2-201) | Biochemical Techniques-II | 1+2=3 | | | | *(BCH-SEC-2-201) | Biochemical Techniques-II | 1+2=3 | |
| | | (AEC-2-201) | English- 2 | 2 | | | | (AEC-2-201) | English- 2 | 2 | |
| | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-1/HCP-2 | 2 | | | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-1/HCP-2 | 2 | |
| THE CANDIDATE HAS TO SELECT MAJOR AND MINOR OUT OF 3 SUBJECTS STUDIED IN THE I AND II SEMESTER AT THE END OF THE SEMESTER II | | | | | | | | | | | |
| THE STUDENT EXITING THE PROGRAMME AFTER ONEYEAR, SECURING48 CREDITS, WILL BE AWARDEDUG CERTIFICATEPROVIDEDHE/SHESECURES 4CREDITSFROMSUMMER INTERNSHIP/COMMUNITY WORK.<br>*CANDIDATE HAS TO SELECT THE SEC OFFERED BY THE DEPARTMENT OF BIOCHEMISTRY IN COLLEGES | | | | | | | | | | | |

## Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## Multidisciplinary Scheme
## Semester I

## Major/Minor

**INSTRUCTIONS FOR PAPER SETTER AND STUDENTS**:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

**(BCH-DSC-1 Maj/Min-101): Biomolecules-I (3+1)**

**Marks: 75 (60 + 15 Int.)**

**Objective:** To learn about biomolecules, their structure and functional significance.

**Learning Outcomes:**

- To understand the chemical and molecular foundations of life and their roles in biological systems.
- To comprehend the classification, structure, chemical properties and biological functions of carbohydrates, and lipids.
- To perform independently the analysis of various biomolecules quantitatively in the laboratory.

### SECTION-I

**(Lectures: 10)**

Introduction to biochemistry and its scope. Water: physical properties, as a biological solvent and structure of water, dissociation of water. pH and pOH, buffer solution. Henderson Hassel Balch equation, acid-base indicators, buffers and physiologically important buffers, dialysis and osmosis.

### SECTION-II

**(Lectures: 10)**

Definition and classification of carbohydrates.Fischer and Haworth structures of carbohydrates.Stereoisomerism and mutarotation.Anomeric forms of monosaccharides. Derivatives of monosaccharides (glycosides, deoxy sugars, amino sugars and other derivatives of biological importance). Oligosaccharides of biological importance (structure of maltose, lactose, sucrose, cellobiose, trehalose, and raffinose). Characteristic reactions of monosaccharides: reactions with hydrazine, hydrogen cyanide, hydroxylamine; reduction and oxidation of sugars; Homo-and hetero-polysaccharides (structures of amylose, amylopectin, starch, glycogen, cellulose, chitin). Hetero-polysaccharides) as components of connective tissue.Polysaccharides of bacterial cell wall.

### SECTION-III

**(Lectures: 10)**

Definition and classification of fatty acids (saturated and unsaturated).Essential fatty acids. Important reactions of functional groups present in fatty acids. Characteristics of fatty acids and fats (saponification, iodine, acid, acetyl and peroxide values).Refractive index, melting point, boiling point and their relation to molecular size.Properties of glycerol.Fats as source of energy.Waxes. Structures, characteristics and functions of lipids: triacylglycerol and phospholipids such as lecithin (Phosphatidylcholines), lysolecithins, cephalins (phosphatidylethanolamines), phosphatidylserines, Phosphatidylinositol, sphingomyelins, plasmalogens, cerebrosides, gangliosides, sulfatides.

**SECTION-IV**

**(Lectures: 10)**

Lipoproteins– composition, classification and biological functions.Liposomes.Terpenes and steroids – Terpenes of biological significance e.g., carotenes, phytol.Cholesterol and other animal sterols.Color reactions of sterols.Sterols of yeast and fungi (mycosterols).Phytosterols.Steroidal hormones.Bile acids. Structure and properties of eicosanoids-prostaglandins, leukotrienes, thromboxane, prostacyclins

# PRACTICALS:

**(BCH-DSC-1 Maj/Min-101): Biomolecules-I**

**Marks: 25 (20 +5 Int.)**

1. Qualitative tests for carbohydrates.
2. Determination of achromatic point of saliva
3. Qualitative tests for Lipids/Fats.
4. Estimation of saponification value of fat/oil.
5. Estimation of acid value of fat/oil.
6. Isolation of starch from potato/gluten from wheat.
7. Demonstration of Osmosis and Dialysis

**Suggested Books:**

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey& D.R. Ferrier, IndianEdition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
4. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principlesof biochemistry (W.H.Freeman,USA)
5. StryerL.,J.M.Berg,J.L.Tymoczko (2023)10th edition,Biochemistry(W.H.Freeman company, NewYork)
6. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry,(Cengage Learning, US)
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998 Principles of Biochemistry volume-1.

## Semester II

## Major/Minor

**INSTRUCTIONS FOR PAPER SETTER AND STUDENTS**:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

**(BCH-DSC-2 Maj/Min-201): Biomolecules-II (3+1)**
**Marks: 75 (60 + 15 Int.)**

**Objective:** To learn about biomolecules, their structure and functional significance.

**Learning Outcomes:**

- To comprehend the structure and functional properties of amino acids and proteins.
- To learn the structure and properties of Nucleic acids and porphyrins.
- To perform independently the analysis of various biomolecules quantitatively in the laboratory.

**SECTION–I**

**(Lectures: 10)**

Amino acids: common structural features. Stereoisomerism and RS system of designating optical isomers. Classification based on the nature of "R" groups. Amino acids present in proteins and non-protein amino acids. Specialized role of amino acids.Physical and chemical properties of amino acids.Titration of amino acids.Rigid and planar nature of a peptide bond.Biologically active peptides.

**SECTION–II**

**(Lectures: 10)**

Proteins: Structural levels Primary, Secondary, Tertiary and Quaternary structures of proteins structure. Folding of peptide chains into regular repeating structures (α helix, β-pleated sheets). β -turn in polypeptides. Forces stabilizing structure and shape of proteins.Native proteins and their conformations.Behaviour of proteins in solutions. Salting in &salting out of proteins. Denaturation of proteins.Structural and functional diversity of proteins, fibrous proteins (keratins, collagen& elastin), globular proteins (hemoglobin, myoglobin) and conjugated proteins.

**SECTION-III**

**(Lectures: 10)**

Nucleic Acids: structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides.Biologically important nucleotides.Double helical model of DNA and forces responsible for it.Different forms of DNA (A, B and Z type), denaturation of DNA.Physical and chemical properties of nucleic acids.Chemical and enzymatic hydrolysis of nucleic acids.Types of RNA and ribozyme.

**SECTION-IV**

**(Lectures: 10)**

Porphyrins: porphyrin nucleus and classification of porphyrins. Metalloproteins, Heme and other metalloporphyrins occurring in nature.Detection of porphyrinsspectrophotometrically and by fluorescence.Chemical nature and physiological significance of bile pigments.

**Suggested Books:**

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey & amp; D.R. Ferrier, Indian Edition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
4. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
5. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)
6. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US)
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998Principles of Biochemistry volume-1

# PRACTICALS:

**(BCH-DSC-2 Maj/Min-201): Biomolecules-II**

**Marks: 25 (20 +5 Int.)**

1. Qualitative tests for amino acids
2. estimation of DNA by Diphenylamine method
3. Separation of amino acids by thin layer chromatography
4. Ammonium sulphate fractionation of proteins.
5. Absorption Spectra of DNA and Proteins by UV visible spectroscopy
6. Estimation of Protein by Biuret /Lowry Method.

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# Bioinformatics

## COURSE OUTLINE & SYLLABUS FOR BIOINFORMATICS

**Students opting for Single major and Single minor(Table 1*)and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)**
**B.Sc., B.Sc. Honours and B.Sc. Honours with Research (1st Year Programme)**

| Sem | Level | Paper Code | Paper Name | Credits (Th+Pr) |
|---|---|---|---|---|
| I | L-100 | BINF-DSC-1 (Maj/Min-101) | Bioinformatics-I | 3+1=4 |
| | | BINF-MDC-1-101 | Introduction to Bioinformatics-I | 2+1=3 |
| | | BINF-SEC-1-101 | Tools for Biological Data Analysis | 0+3=3 |
| II | L-100 | BINF-DSC-2 (Maj/Min-201) | Bioinformatics-II | 3+1=4 |
| | | BINF-MDC-2-201 | Introduction to Bioinformatics-II | 2+1=3 |
| | | BINF-SEC-2-201 | Computational Tools for Small and Macromolecules | 0+3=3 |

**Notes: -**

1. MDC/IDC subject is offered by The Department of Bioinformatics for the students of allied/other streams. The students opting for Bioinformatics as DSC subject will select MDC/IDC from the list of courses offered by the allied/other departments.
2. Language paper chosen as MIL-1, MIL-2, MIL-3 and MIL-4 should be in the same subject.

## SEMESTER - I

### Paper: BINF-DSC-1-(Maj/Min-101): BIOINFORMATICS-I

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

**Course Duration:**
1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

**Objective:**
Introduction to Biomolecules exposes the students to study structure and properties of biomolecules and their related bioinformatics resources.

**General Instruction:**
- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

### UNIT – I

**Nucleic acids:**
Nitrogenous Bases, nucleosides and nucleotides.

**Overview of DNA structure**: Watson and Crick DNA double helix model, Conformations of DNA: A-DNA, B-DNA & Z-DNA.

**Overview of RNA**: types and roles of different RNAs: mRNA, rRNA, tRNA, snRNA and snoRNA.

**Amino acids:** Standard amino acids and the general properties.

**Classification and structures**: non-polar, polar but uncharged and charged amino acids.
**Introduction to Proteins:**Overview ofProtein Structure.
Primary structure - definition, peptide bond, dihedral angles, and Ramachandran plot.
Secondary structure - definition, alpha helix, beta sheet, and beta turn.
Tertiary structure - definition, interactions stabilizing tertiary structure.
QuaternaryStructure - definition, interactions stabilizing quaternary structure

## UNIT – II

**Introduction to Bioinformatics:** Definitions, History, Goal, Scope and its Applications.

**Literature Database:**PubMed.

**Biological Databases Classification:** Primary, Secondary and Specialized databases.
**Nucleotide Sequence Database:**Genbank, EMBL, INSDC, Indian Biological Data Centre (IBDC) and its data portals.
**Protein Sequence Database:**Uniprot.
**Database flat file formats:**Genbank/Genpept, EMBL and Uniprot flat files.
**Sequence file formats:** FASTA

**Sequence Alignment:**Evolutionary basis of Alignment, Concept of Sequence Homology, Similarity, and Identity.Applications of Sequence Alignments.

**Pairwise sequence alignment:** Concept of Global and Local alignment, Dot Matrix method, Dynamic programming (Needleman Wunsch algorithm and Smith-Waterman algorithm) Concepts behind General gap and affine gap penalty.

**Recommended Books:**

1. Donald Voet and Judith G. Voet (2010) Biochemistry, 4th Edition, Wiley Publisher.
2. David L. Nelson and Michael M. Cox (2017). Lehninger Principles of Biochemistry, 7th Edition, WH Freeman Publisher.
3. Reginald H. Garrett and Charles M. Grisham (2023) Biochemistry, 7th Edition, Brooks/Cole Publisher.
4. Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
5. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
6. Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.

**BINF-DSC-1-(Maj/Min-101):BIOINFORMATICS-I (PRACTICAL)**

External Practical Marks: 20
Internal Practical Marks: 05
Total Practical Marks: 25

1. Draw the amino acids using Chemsketch.
2. Plotting of Ramachandran map of a protein.
3. Visualization of DNA using Pymol.
4. Searching literatures over PubMed.
5. Understanding and using Sequence information resources on web: EMBL-ENA,Genbank, EMBL, IBDC and Uniprot.
6. The information retrieval using Gquery.
7. Performing Sequence alignment using DOT PLOT method

## SEMESTER - II

**Paper: BINF-DSC-2-(Maj/Min-201): BIOINFORMATICS-II**

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

**Course Duration:**
1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

**Objective:**
This subject offers various bioinformatics tools that helps in analysis of biological sequence data and phylogenetic analysis of proteins and nucleotides by using various computational tools.

**General Instruction:**
- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

### UNIT – I

**Tools for Pairwise Sequence Alignment:**BLAST: Algorithm, types, understanding the BLAST output and the significance of E-value and Bit score, different uses of BLAST.

FASTA algorithm, Difference between BLAST & FASTA.

**Substitution matrices:**Amino acid substitution matrices (PAM and BLOSUM)

**Multiple Sequence Alignment (MSA):** SP (Sum of Pairs) measure to determine scoring function, Multiple Sequence Alignment methods: Progressive (Clustal Omega and PILEUP), Iterative and Block-based methods of MSA. Application of MSA.

**Tools for MSA:**Clustal Omega, Clustal-X and T-coffee.

### UNIT – II

**Phylogenetic Analysis:** Basic terminology in Phylogenetics, Introduction to phylogenetic analysis and its application, Phylogenetic tree terminology and topologies.

**Methods of Phylogenetic Tree Construction:**
Distance Based Methods: NJ, FM & UPGMA
Character Based Methods: Maximum Parsimony & Maximum Likelihood.

Phylogenetic Tree Evaluation Methods: Bootstrapping & Jackknifing.

**Tools for Phylogenetic Analysis:** Introduction to Phylip and MEGA packages.

**Protein Structural databases:**PDB, SCOP and CATH

**Recommended Books:**


- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformatics: Ardpractical guide to the analysis of genes and proteins. Ed. By Baxevanis 3rd Edition, (1998) John Wiley & Sons, Inc. Publication
- BLAST by Ian Korf, Mark Yandell, Joseph Bedell, O'Reilly, 1st Edition, (2003).
- Molecular Evolution: a phylogenetic approach, (1998), 352p, ROM and Holmas FC, Blackwell Science.


**BINF-DSC-2-(Maj/Min-201):BIOINFORMATICS-II (PRACTICAL)**

| | |
|---|---|
| External Practical Marks: | 20 |
| Internal Practical Marks: | 05 |
| Total Practical Marks: | 25 |

1. Performing BLASTp/n and interpreting its results.
2. Performing Multiple Sequence Alignment using Clustal Omega
3. Building Phylogenetic tree using:
   a) Distance-based & character-based methods by PHYLIP or MEGA
   b) Tree evaluation
4. Understanding the PDB and its file format
5. Downloading and visualizing protein structure from PDB

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# Biotechnology

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

| Table 1* | | | | | | *Table 2** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | Total Credits | Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | *Total Credits* |
| | | | | | | | | | | | |
| I | L-100 | BIOT-DSC-(MAJ/MIN)-1-101 | Introduction to Biotechnology | 3+1 | 24 | I | L-100 | BIOT-DSC-1 (MAJ/MIN)-101 | Introduction to Biotechnology | 3 + 1 | *24* |
| | | *MIN B-1 | | 3+1 | | | | Major | | 3+1 | |
| | | *MIN C-1 | | 3+1 | | | | Major | | 3+1 | |
| | | BIOT-MDC-1- 101 | Foundations of Biotechnology-I | 2+1 | | | | BIOT-MDC1- 101 | MDC | 2 + 1 | |
| | | **BIOT-SEC-1-101 | Biosafety & Biosecurity | 2+1 | | | | VAC-1-101 | Value-Added Course | 2 | |
| | | BIOT-VAC-1-101 | Value Added Course | 2 | | | | SEC-1-101 | SEC | 2+1 | |
| | | AEC-1-101 | Eng-1 | 2 | | | | AEC-1-101 | Eng-1 | 2 | |
| | | BIOT-PUN (Comp-1)-101<br>BIOT-HCP (Comp-1)-101 | Pbi/HCP | 2 | | | | PUN (Comp-1)-101<br>HCP (Comp-1)-101 | Pbi/HCP | 2 | |
| II | L-100 | BIOT-DSC-(MAJ/MIN)-2-201 | Foundations of Microbiology | 3+1 | 24 | II | L-100 | BIOT-DSC-2 (MAJ/MIN)-201 | Foundations of Microbiology | 3 + 1 | *24* |
| | | *MIN-B-2 | | 3+1 | | | | Major | | 3+1 | |
| | | *MIN-C-2 | | 3+1 | | | | Major | | 3+1 | |
| | | BIOT-MDC-2- 201 | Foundations of Biotechnology-II | 2+1 | | | | MDC-2-201 | MDC | 2 + 1 | |
| | | **BIOT-SEC-2-201 | Biostatistics | 2+1 | | | | VAC-2-201 | Value-Added Course | 2 | |
| | | AEC-2-201 | Eng-2 | 2 | | | | SEC-2-201 | SEC | 2+1 | |
| | | BIOT-PUN (Comp-2)-201<br>BIOT-HCP (Comp-2)-201 | Pbi/HCP | 2 | | | | AEC-2-201 | Eng-2 | 2 | |
| | | BIOT-VAC-2-201 | Value Added Course | 2 | | | | PUN (Comp-2)-201<br>HCP (Comp-2)-201 | Pbi/HCP | 2 | |

*The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester*

*The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work.*

## Semester-I

## BIOT-DSC-(MAJ/MIN)-1-101: INTRODUCTION TO BIOTECHNOLOGY

**Learning Outcomes:** This course will introduce the basic concepts and skills of biotechnology to the students. They will learn about the history of biotechnology; the foundations of modern biotechnology; the role of biotechnology in industry, environment and modern medicine and the ethical implications of biotechnology.

**Course Duration**
45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

**Credits : 4 credits** (Each credit is equal to 25 marks)

| External Theory | Internal Theory | External practical | Internal practical |
|---|---|---|---|
| 68 | 07 | 22 | 03 |

**Theory Paper:** The exam of this course will be of 3 hours duration.

**Instructions for paper setters and candidates**:

The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

**Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

### UNIT I

Advent, scope and basics of biotechnology
Bacteria as work horses of biotechnology; *E.coli* as the model bacterium
*Saccharomyces cerevisiae* in Biotechnology
Introduction of multicellular organisms as research models:*Drosophila melanogaster, Caenorhabditiselegans, Daniorerio, Musmusculus,, Arabidopsis thaliana* as model for plant genetics.

### UNIT II

Genome organization in prokaryotes and eukaryotes, Molecular nature of the genetic material.
Introduction of concept and applications of genomics, transcriptomics, proteomics and metabolomics.
Introduction to Principle and basic techniques like sterilization, centrifugation&Agarose gel electrophoresis.

### UNIT III

Applications of biotechnology: Green technology to controlpollution. Role of biotechnology in medicine and diagnostics, agriculture and industry.Introduction of gene therapy.

### UNIT IV

Genetically modified organisms (GMOs)-transgenic plants and animals and their applications in biotechnology.Bioterrorism and biowarfare.Ethical, Social and Legal implications of Biotechnology.

**Reference Books:**

1. Comprehensive Biotechnology, Murray Moo-Young, 2ndedition(2011), Pergamon Press.
2. Introduction to Biotechnology ,William J. Thieman& Michael A. Palladino, 3rd edition (2012),Benjamin Cummings.
3. Biotechnology Expanding Horizons, B D Singh,4$^{th}$edition(2012),Kalyani Publishers.
4. The Ethics of Biotechnology (Biotechnology in the21st Century), Jonathan Morris, Chelsea HousePub (L),1stedition(2005).
5. Biotechnology, Applying the Genetic Revolution, David P.Clark& Nanette J.Pazdernik,1stedition(2008), Academic Cell.
6. Molecular Biotechnology, Sandy B.Primrose, 2$^{nd}$edition(1991), Blackwell Scientific Publications,
7. Biotechnology:Demystifying the concepts, David Bourgaize,Thomas R. Jewell & Rodolfo G. Buiser,1$^{st}$ Edition(1999), Benjamin Cummings.
8. Ethical Issues in Biotechnology, Richard Sherlock & John D. Morrey, 1st edition (2002), Rowman& Little field Publishers.

**BIOT-DSC-(MAJ/MIN)-1-101: INTRODUCTION TO BIOTECHNOLOGY(Practicals)**

1. Introduction and use of Auto-pipettes, pH meter, spectrophotmeter
2. Use of centrifuges, light microscope, electrophoretic apparatus
3. Introduction to vortex mixer, magnetic stirrer, rocker, laminar hoods, autoclave
4. DNA isolation
5. Restriction digestion
6. Cell lysis of bacterial cell (Lysozyme digestion)

## Semester-II

## BIOT-DSC-(MAJ/MIN)-2-201: FOUNDATIONS OF MICROBIOLOGY

**Learning Outcomes:** This paper exploresthe interactions of the Microbes with environment and other living organisms. This will enhance the student's understanding about how the microbes play a significant role in medical science and industries. The study of microbesfrom basic to advance level, with an understanding of biochemistry,cell structure, and application makesthis paper significant.

**Course Duration**

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

**Credits: 4 credits** (Each credit is equal to 25 marks

| External Theory | Internal Theory | External practical | Internal practical |
|---|---|---|---|
| 68 | 07 | 22 | 03 |

**Theory Paper:** The exam of this course will be of 3 hours duration.

**Instructions for paper setters and candidates**:
The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.
**Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each**

### UNIT I

History of Microbiology: A. Leewenhook, L. Pasteur, R. Koch, J. Lister, J. Tyndall,Alexander Flemming, Koch postulates.
Principle and applications of microscopy: Bright field, dark field, phase contrast, fluorescent, electronmicroscopy.

### UNIT II

Microbial classification; Conventional and Modern techniques.Morphology of bacteria with major emphasis on bacterial structure especially cell wall.Gram-positive and Gram-negative bacteria. Microbial spores.

### UNIT III

Microbial growth, phases of growth, generation time, growth rate, monoauxic, diauxic andsynchronous growth. Chemostat and turbidostat.
Physical and chemical agents to kill microbes major emphasis on sterilization process.

### UNIT IV

Normal micro flora in human.Types of microbial pathogens and diseases caused by them.
Microbial interactions like symbiosis and antibiosis.

**Reference Books:**

1. Cappuccino J.G., Sherman N. (2007). Microbiology: A laboratory manual, (PearsonBenjamin Cummings).
2. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction(Benjamin/ Cummings publishingcompany, Inc).
3. R. Y. Stanier, M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillianpress London).
4. M.J. Pelczar, E.C. Sun Chan, N.R. Krieg (1986). Microbiology (Tata McGraw HillPublication, New
Delhi).
5. H.G. Schlegel, C.Zaborosch, M. Kogut (1993).General microbiology (CambridgeUniversity Press).
6. S.C. Prescott, C.G. Dunn (1959). Industrial microbiology (McGraw- Hill).
7. Purohit, S.S. (2003). Microbiology: Fundamentals and applications (Agrobios, India).
8. Postgate, J.R. (2000). Microbes and man (Cambridge University Press).

## BIOT-DSC-(MAJ/MIN)-2-201: FOUNDATIONS OF MICROBIOLOGY (Practicals)

1. Cleaning of glass wares, preparation of media, cotton plugging and sterilization.
2. Isolation of microorganisms from air, water and soil samples.
3. Dilution and pour plating techniques.
4. Gram staining
5. Growth curve of microorganisms.

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# Botany

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)**
**and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)**
**B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BOTANY**

| Table 1* | | | | | | Table 2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** |
| I | L-100 | (BOT-DSC-1 Maj/Min-101) | Plant Diversity I | 3+1=4 (x3) | 24 | I | L-100 | (BOT-DSC-1 Maj/Min-101) | Plant Diversity I | 3+1=4 (x3) | 24 |
| | | (BOT-MDC-1-101) | Plants–Importance and Propagation | 3 | | | | (BOT-MDC-1-101) | Plants–Importance and Propagation | 3 | |
| | | (BOT-VAC-1-101) | Nursery and Gardening | 2 | | | | (BOT-VAC-1-101) | Nursery and Gardening | 2 | |
| | | (BOT-SEC-1-101) | Basic Lab and Field Skills in Botany | 2+1=3 | | | | (BOT-SEC-1-101) | Basic Lab and Field Skills in Botany | 2+1=3 | |
| | | (AEC-1-101) | Eng-1 | 2 | | | | (AEC-1-101) | Eng-1 | 2 | |
| | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | | | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (BOT-DSC-2 Maj/Min-201) | Plant Diversity II | 3+1=4 (x3) | 24 | II | L-100 | (BOT-DSC-2 Maj/Min-201) | Plant Diversity II | 3+1=4 (x3) | 24 |
| | | (BOT-MDC-2-201) | Utilization and Conservation of Plants | 3 | | | | (BOT-MDC-2-201) | Utilization and Conservation of Plants | 3 | |
| | | (BOT-VAC-2-201) | Biodiversity and Conservation | 2 | | | | (BOT-VAC-2-201) | Biodiversity and Conservation | 2 | |
| | | (BOT-SEC-2-201) | Biofertilizers | 2+1=3 | | | | (BOT-SEC-2-201) | Biofertilizers | 2+1=3 | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | (AEC-2-201) | Eng-2 | 2 | |
| | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | | | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II | | | | | | | | | | | |
| The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. | | | | | | | | | | | |

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## (SEMESTER – I)

**Course Title: PLANT DIVERSITY-I** **Course Code: BOT-DSC-1 Maj/Min-101**

**Learning Outcome:** The basic objective of this paper is to make students aware of the diversity in life forms of plant kingdom. A systematic study of algae, fungi and bryophytes would familiarize students with the structural differentiation and heterotrophic and autotrophic modes of nutrition in plant kingdom. This paper will form the basis of advanced study in Botany.

**Max. Marks: 100** **Course Credits =** 3+1=4

| | | | |
|---|---|---|---|
| 1. | Theory: 75 | Written Exam: 67 | Internal Assessment: 8 |
| 2. | Practical: 25 | Written Exam: 22 | Internal Assessment: 3 |

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

**Instructions for Paper Setters:**
Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

### UNIT – I 12 Hours

Algae: General Characters; systematic position, structure and life history of *Oscillatoria*(Cyanophyceae) *Volvox* (Chlorophyceae), *Ectocarpu*s (Phaeophyceae), *Batracospermum* (Rhodophyceae) and economic importance of algae.

### UNIT – II 11 Hours

Fungi: General characters; systematic position, structure and life history of *Albugocandida* (White rust of crucifers), *Rhizopus* and *Saccharomyces*.

### UNIT – III 11 Hours

Systematic position, structure and life history of *Ustilagotritici* (Smut of wheat), *Colletotrichumfalcatum* (Red rot of sugarcane).
Symbiotic Association: General account of Lichens and their significance.

### UNIT – IV 11 Hours

Bryophyta: General characters; systematic position, structure, reproduction and life cycle of *Riccia*(Hepaticopsida); *Anthoceros*(Anthocerotopsida) and *Funaria* (Bryopsida) excluding developmental stages.

**Suggested Readings:**

1. Alexopoulos, C. J. Mims, C. W`. and Blackwell, M. 1996. Introductory Mycology. John Wiley and Sons, Inc. USA.
2. Dube, H.C. 1990. An Introduction to Fungi. Vikas Publishing House Pvt. Ltd., Delhi.
3. Goyal, J. P. Dhand, Neelam and Saini, Aruna. 2011. Foundations of Botany. Trueman Book Company Jalandhar.
4. Pandey, B.P. 2009. Botany for Degree Students. S. Chand & Co. Ltd., New Delhi.
5. Sharma, O.P. 1992. Textbook of Thallophytes, McMillan India Ltd., New Delhi.
6. Sharma, P.D. 1991. The Fungi. Rastogi& Co., Meerut.
7. Smith, G.M. 1971. Cryptogamic Botany. Vol. 1, Algae & Fungi, Tata McGraw Hill Publishing Co. New Delhi.
8. Singh, V. Pandey, P.C. and Jain, D.K. 2012. Textbook of Botany, Diversity of Microbes and Cryptogams. Rastogi Publications, Meerut & New Delhi.
9. Srivastava, H.N. 2013. Pradeep's Botany Vol. I (Diversity of Microbes and Cryptogams), Pradeep Publications, Jalandhar (India).
10. Vashishta, B. R. 1999. Botany for Degree Students. Algae. S. Chand and Company Ltd., New Delhi.
11. Vashishta, B. R. 1999. Botany for Degree Students. Fungi. S. Chand and Company Ltd., New Delhi.
12. Puri, P. 1980. Bryophyta. Atma Ram & Sons, Delhi.
13. Vashishta, B. R. 1999. Botany for Degree Students. Bryophyta. S. Chand and Company Ltd., New Delhi

**Suggested laboratory exercises:**

1. Study of morphology of various genera included in algae and fungi.
2. Study of Crustose, Foliose and Fructicose types of Lichen thalli.
3. Histopathological study of White rust of crucifers, Smut of wheat and Red-rot of sugarcane.
4. *Riccia*: Study of morphology of thallus, Study of L.S. Mature sporogonium (through permanent slides) and Preparation of permanent stained slide of V.S. *Riccia*thallus.
5. *Anthoceros*: Study of morphology of thallus, Study of T.S. Thallus passing through antheridia, T.S. Thallus passing through archegonia, L.S. Mature sporogonium. (through permanent slides) and Preparation of permanent stained slide of V.S. thallus.
6. *Funaria*: Study of morphology, Study of L.S. Male receptacle, L.S. Female receptacle, L.S. Capsule (through permanent slides) and Preparation of permanent stained slide of T.S. of stem.

## GUIDELINES FOR PRACTICAL EXAMINATION

**Time- 3 Hrs** **Max Marks: 25**
**Practical: 22** **Internal Assessment: 03**

1. Identify and write illustrated morphological notes on specimen A and B. (06)
2. Cut T.S., stain and make a permanent mount of specimen C (from bryophytes). Identify and draw its labelled diagram and show slide to examiner. (06)
3. Identify the slides D and E giving at least one reason for each. (04)
4. Practical Note-book (03)
5. Viva-voce. (03)

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## SEMESTER – II

**Course Title: PLANT DIVERSITY-II** **Course Code: BOT-DSC-2 Maj/Min-201**

**Learning Outcome:** The basic objective of this paper is to make the students aware of the diversity in various life forms of plant kingdom. A sequential study ranging from Pteridophytes - the first vascular land plants to gymnosperms, would enable students to have a broad prospective of evolutionary trends in plant kingdom.

**Max. Marks: 100** **Course Credits =** 3+1=4

1. Theory: 75 Written Exam: 67 Internal Assessment: 8
2. Practical: 25 Written Exam: 22 Internal Assessment: 3

**Periods per week:** 1. Theory- Four of 45 minutes duration (Three Hours)
2. Practical- One of two hours duration (Two Hours)

**Instructions for Paper Setters:**

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

### UNIT – I 12 Hours

Pteridophyta: General characters; systematic position, structure and reproduction of *Rhynia* (Psilophytopsida) and *Selaginella* (Lycopsida) excluding developmental stages, heterospory and seed habit.

### UNIT – II 11 Hours

Systematic position, structure and reproduction of *Equisetum* (Sphenopsida) and *Pteris* (Pteropsida) excluding developmental stages.

### UNIT – III 11 Hours

Gymnosperms: General characters and economic importance of gymnosperms; systematic position, structure and reproduction of*Cycas* excluding developmental stages.

### UNIT – IV 11 Hours

Systematic position, structure and reproduction of *Pinus* and *Ephedra* excluding developmental stages.

**Suggested Readings:**

1. Goyal, J. P. Dhand, Neelam and Saini, Aruna 2011. Foundations of Botany. Trueman Book Company Jalandhar.
2. Pandey, B.P. 2009. Botany for Degree Students. S. Chand & Co. Ltd., New Delhi
3. Sharma, O.P. 1990. Textbook of Pteridophyta, McGraw Hill Publishing Co., New Delhi.
4. Singh, V. Pandey, P.C. and Jain, D.K. 2012. Textbook of Botany, Diversity of Microbes and Cryptogams. Rastogi Publications, Meerut & New Delhi.
5. Smith, G.M. 1971. Cryptogamic Botany. Vol. II, Bryophytes and Pteridophytes, Tata McGraw Hill Publishing Co., New Delhi.
6. Srivastava, H.N. 2013.Pradeep's Botany Vol. I (Diversity of Microbes and Cryptogams), Pradeep Publications, Jalandhar (India).
7. Bhatnagar, S.P. and Moitra, A. (1996) *Gymnosperms*. New Age International (P) Ltd. Publishers, New Delhi, India.
8. Chopra, G.L. Textbook of Gymnosperms. S. Nagin. Delhi.1976
9. Sporne, K.R. The morphology of Gymnosperms, Hutichnson& Co. (Publishers) Ltd, London 1965.

**Suggested laboratory exercises:**

1. *Selaginella*: Study of morphology, study of L.S. sporangiferous spike (through permanent slide) and preparation of permanent slide of T.S. stem.
2. *Equisetum*: Study of morphology, study of L.S. strobilus and T.S. strobilus (through permanent slide) and preparation of permanent slide of T.S. Aerial stem passing through internode.
3. *Pteris*: Study of morphology, study of mature prothallus (through permanent slide) and preparation of permanent slide of T.S. petiole.
4. *Cycas*:Morphology (coralloid root, scale and foliage leaves, male cone, megasporophyll). Preparation of permanent slide of *Cycas* (T.S. leaflet)
5. *Pinus*:Morphology (dwarf shoot, male and female cone and seed), preparation of permanent slide of *Pinus* (T.S. Needle).
6. *Ephedra*:Morphology (male and female cone), preparation of permanent slide of *Ephedra* (T.S. Stem).

## GUIDELINES FOR PRACTICAL EXAMINATION

**Time: 3 Hrs** **Max Marks: 25**
**Practical: 22** **Internal Assessment: 03**

1. Identify and write illustrated morphological note on specimen A and B. (06)
2. Cut T.S., stain and make a permanent mount of specimen C. Identify and draw its labelled diagram and show slide to examiner. (06)
3. Identify the slides D and E giving at least one reason for each. (04)
4. Practical Note- book (03)
5. Viva-Voce. (03)

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# Chemistry

## CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MAJOR SUBJECT AS PER TABLE 1 AND TABLE 2

| Table 1 | | | | | | Table 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Total Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Total Credits** |
| I | L-100 | (CHM-DSC-1 MAJ/MIN-101) | GENERAL CHEMISTRY – 1 | 4 | 24 | I | L-100 | (CHM-DSC-1 MAJ/MIN-101) | GENERAL CHEMISTRY – 1 | 4 | 24 |
| | | B1 | To be chosen by the student | 4 | | | | B1 | To be chosen by the student | 4 | |
| | | C1 | To be chosen by the student | 4 | | | | C1 | To be chosen by the student | 4 | |
| | | IDC-1 | To be chosen from the available options | 3 | | | | IDC-1 | To be chosen from the available options | 3 | |
| | | (CHM-SEC-1-101) | Hands on chemistry Lab skills-1/ to be chosen from available options | 3 | | | | (CHM-SEC-1-101) | Hands on chemistry Lab skills-1/ to be chosen from available options | 3 | |
| | | ENG-1 | Compulsory | 2 | | | | ENG-1 | Compulsory | 2 | |
| | | VAC-1 | To be chosen from the common basket | 2 | | | | VAC-1 | To be chosen from the common basket | 2 | |
| | | PBI-1/HCP-1 | Compulsory | 2 | | | | PBI-1/HCP-1 | Compulsory | 2 | |
| II | L-100 | (CHM-DSC-2 MAJ/MIN-201) | GENERAL CHEMISTRY – 2 | 4 | 24 | II | L-100 | (CHM-DSC-2 MAJ/MIN-201) | GENERAL CHEMISTRY – 2 | 4 | 24 |
| | | B2 | To be chosen by the student | 4 | | | | B2 | To be chosen by the student | 4 | |
| | | C2 | To be chosen by the student | 4 | | | | C2 | To be chosen by the student | 4 | |
| | | IDC-2 | To be chosen from the available options | 3 | | | | IDC-2 | To be chosen from the available options | 3 | |
| | | (CHM-SEC-2-201) | Hands on chemistry Lab skills-2/ to be chosen from available options | 3 | | | | (CHM-SEC-2-201) | Hands on chemistry Lab skills-2/ to be chosen from available options | 3 | |
| | | ENG-2 | Compulsory | 2 | | | | ENG-2 | Compulsory | 2 | |
| | | VAC-2 | To be chosen from the common basket | 2 | | | | VAC-2 | To be chosen from the common basket | 2 | |
| | | PBI-2/HCP-2 | Compulsory | 2 | | | | PBI-2/HCP-2 | Compulsory | 2 | |

## CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MINOR SUBJECT AS PER TABLE 1 AND TABLE 2

| Table 1 | | | | | Table 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** |
| I | L-100 | (CHM-DSC-1 MAJ/MIN-101) | GENERAL CHEMISTRY – 1 | 4 | I | L-100 | (CHM-DSC-1 MAJ/MIN-101) | GENERAL CHEMISTRY – 1 | 4 |
| II | L-100 | (CHM-DSC-2 MAJ/MIN-201) | GENERAL CHEMISTRY – 2 | 4 | II | L-200 | (CHM-DSC-2 MAJ/MIN-201) | GENERAL CHEMISTRY – 2 | 4 |
| III | L-200 | (CHM-DSC-3 MAJ/MIN-301) | GENERAL CHEMISTRY – 3 | 4 | III | L-200 | (CHM-DSC-3 MAJ/MIN-301) | GENERAL CHEMISTRY – 3 | 4 |
| | L-200 | (CHM-DSC-5-MIN-302/MAJ-401) | GENERAL CHEMISTRY – 4 | 4 | | L-200 | (CHM-DSC-5-MIN-302/MAJ-401) | GENERAL CHEMISTRY – 4 | 4 |
| IV | L-200 | (CHM-DSC-6 MIN-401/MAJ-402) | GENERAL CHEMISTRY – 5 | 4 | IV | L-200 | (CHM-DSC-6 MIN-401/MAJ-402) | GENERAL CHEMISTRY – 5 | 4 |

| CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS IDC AS PER TABLE 1AND TABLE 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Table 1** | | | | | **Table 2** | | | | |
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** |
| I | L-100 | (CHM-IDC-1-101) | Basic Analytical Methods-1 | 3 | I | L-100 | (CHM-IDC-1-101) | Basic Analytical Methods-1 | 3 |
| II | L-100 | (CHM-IDC-2-201) | Basic Analytical Methods-2 | 3 | II | L-100 | (CHM-IDC-2-201) | Basic Analytical Methods-2 | 3 |
| III | L-200 | (CHM-IDC-3-301) | Colours and Dyes | 3 | III | L-200 | (CHM-IDC-3-301) | Colours and Dyes | 3 |

| CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS VAC AS PER TABLE 1 AND TABLE 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Table 1** | | | | | **Table 2** | | | | |
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** |
| I | L-100 | (CHM-VAC-1-101) | Understanding Science-I | 2 | I | L-100 | (CHM-VAC-1-101) | Understanding Science-I | 2 |
| II | L-100 | (CHM-VAC-2-201) | Understanding Science-II | 2 | II | L-100 | (CHM- VAC -2-201) | Understanding Science-II | 2 |
| IV | L-200 | (CHM-VAC-3-301) | Food poisons | 2 | IV | L-200 | (CHM- VAC -3-301) | Food poisons | 2 |

**Note:-**

1. In the first semester, studentsare offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

| | |
|---|---|
| **Semester** | **I** |
| **Code of Paper** | **CHM-DSC-1-MAJ/MIN-101** |
| **Title of Paper** | **General Chemistry-1** |
| **Total Credit** | **04 (03 Theory + 01 Practical)** |
| **Total Marks** | **100(75Theory + 25 Practical)** |
| **Marks distribution of the Theory paper** | **End semester = 67**<br>**Internal Marks = 08** |
| **Marks distribution of the Practical paper** | **End semester = 22**<br>**Internal Marks = 03** |
| **Contact Hours** | **Theory: 3 Hrs. per week**<br>**Practical: 2 Hrs. per week** |

**Applicable to the students**

1. **Those who have opted for chemistry as a Major subject in Table 1 or Table 2**
2. **Those who have opted for chemistry asaMinor subject in Table 1 or Table 2**

**LEARNING OUTCOMES SPECIFIC TO DISCIPLINE**

By studying this course, the students will be able to:

- Explain the electronic configuration of elements & ions based on different principles and rules.
- Explain the chemical behaviour of different elements from their position in the periodic table.
- Explain the differential behaviour of organic compounds based on fundamental concepts learnt.
- Explain and differentiate between various types of reagents, organic reactions, reactive intermediates and how to assign a formal charge to an ionic species.
- The student shall also be able to differentiate between geometrical and conformational isomers.
- Explain the factors influencing the rate of reactions based on various theories of chemical kinetics.
- Explain the concept of the order of a reaction by different methods.
- Understand different types of catalysts and the mechanism of enzyme catalysis.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

**GENERIC OUTCOMES**

The first year of the course is designed to inculcate a scientific temper amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out analysis, evaluation and draw logical conclusions.To hone their critical thinking and enhance creativity, which shall lead to innovative and out-of-the-box thinking, which shall help in complex problem-solving situations in higher classes.They are also encouraged to work and improve upon their communication skills, through proper listening, literature survey and expression through word and print.

**Semester-I**
**Syllabus for Theory Paper**
**CHM-DSC-1-MAJ/MIN-101**
**General Chemistry-1**

| | |
|---|---|
| **End semester = 67** | **Credits - 03** |
| **Internal Marks = 08** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

## UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

### Atomic Structure (8 Hrs.)

Idea of de-Broglie matter waves, Heisenberg uncertainty principle, atomic orbitals, Schrodinger wave equation, the significance of $\psi$ and $\psi^2$, quantum numbers, radial and angular wave functions (excluding mathematical relations), probability distribution curves, shapes of *s*, *p*, and *d*orbitals. Aufbau and Pauli exclusion principles, Hund's multiplicity rule, Electronic configurations of the elements and ions.

### Periodic Properties (7 Hrs.)

Position of elements in the periodic table, effective nuclear charge and its calculations, Atomic and ionic radii, ionisation energy, electron affinity and electronegativity– definition, methods of determination or evaluation, trends in the periodic table and applications in predicting and explaining the chemical behaviour.

## UNIT II: ORGANICCHEMISTRY(15 Hrs.)

### Structure and Bonding (5 Hrs.)

Hybridisation, Bond lengths and bond Angles, Bond Energy, Localised and Delocalized chemical bond, Van der Waal's interaction, Resonance and Resonance effect, Hyperconjugation, Inductive and Field effect, Electromeric effect, Hydrogen Bonding.

### Reactive Intermediates (5 Hrs.)

Curved arrow notation, Drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking, Types of Reagents – Electrophiles and nucleophiles, Types of Organic Reactions.Reactive intermediates-Carbocations, Carbanions, Free Radicals, Carbenes, Arynes and Nitrenes (with examples). Assigning Formal charges on intermediates and other ionic species

### Geometrical and Conformational Isomerism (5 Hrs.)

### Geometrical isomerism

Cause and conditions for geometrical isomerism, Nomenclature of geometrical isomers- cis and trans, E and Z system, Determination of configuration of geometrical isomers.

### Conformational Isomerism

Representation of conformations-Sawhorse and Newman formulae, Conformational analysis of ethane, propane, n-butane, cyclohexane, Equatorial and Axial bonds.Conformations of Mono and di-substituted derivatives of cyclohexanes.

## UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

### Basic Concepts of Mathematics (3Hrs.)

Logarithmic relations, differentiation and integration of functions like $e^x$, $x^n$, sin x, and log x. Terms of mean and median, precision and accuracy in chemical analysis.

### Chemical Kinetics-I(6 Hrs.)

Chemical kinetics and its scope, rate of a reaction, factors influencing the rate of a reaction-concentration,temperature, pressure, solvent, light, and catalyst. Concentration dependence of rates,mathematical characteristicsof simple chemical reactions – zero order, first order, second order, pseudo order, half-life, and mean life.Determination of the order of reaction – differential method, method of integration,method of half-life periodandisolation method, Radioactivedecayasafirst-orderphenomenon.

### Chemical Kinetics-II(6 Hrs.)

Theories of Chemical Kinetics: Effect of temperature on rate of reaction, Arrhenius equation, concept of activation energy, Simple collision theory based on hard sphere model, transition state theory (equilibrium hypothesis), Expression for the rate constant based on equilibrium constant and thermodynamic aspects.

Catalysis and general characteristics of catalytic reactions,Homogeneous catalysis, acid-base catalysis and enzyme catalysis, including their mechanisms, Michaelis-Menten equation for enzyme catalysis and its mechanism.

### Books/Literature suggested

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., Basic Inorganic Chemistry; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., Concise Inorganic Chemistry; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., Inorganic Chemistry; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield, W.W., Wesky, A., Inorganic Chemistry; Pubs: Addison-Wesley Publishing Company, 1984.
6. Miessler, G.L., Tarr, D.A., Inorganic Chemistry; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Puri, B.R., Sharma, L.R., Kalia, K.K., Principles of Inorganic Chemistry; 30th edition, Pubs: Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.

15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson Education, 2008.
20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

## Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

## Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

**Semester-1**
**Syllabus for Practical Paper**
**CHM-DSC-1-MAJ/MIN-101**
**General Chemistry-1**

| | |
|---|---|
| **End semester = 22** | **Credit - 01** |
| **Internal Marks = 03** | **Contact Hours – 2/week** |
| **Examination Time allowed** | **3 Hours** |

## I. INORGANIC CHEMISTRY

1. Preparation of standard solutions.
2. Acid-base titrations
    a. HClvsNaOH
    b. $Na_2CO_3$vsHCl
    c. Mixture of bases versus HCl

## II.ORGANIC CHEMISTRY

3. Calibration of the thermometer: 80-82$^o$ (Naphthalene), 113-114$^o$C(acetanilide),132.5-133$^o$C(Urea), 100$^o$C (distilled Water)
4. Determination of melting pointsof some organic compounds: Naphthalene 80-82$^o$, Benzoic acid 121.5-122$^o$ Urea, 132.5-133$^o$, Cinnamic acid 132.5-133$^{o,}$ Acetanilide 113-5-114$^o$, p-Dichlorobenzene 52$^o$. Aspirin135$^o$.

## III. PHYSICAL CHEMISTRY

5. Viscosity: To determine the viscosity of Brine Solution(20%), n-Butylalcohol, cyclohexane

### Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

| | |
|---|---|
| **Semester** | **II** |
| **Code of Paper** | **CHM-DSC-2-MAJ/MIN-201** |
| **Title of Paper** | **General Chemistry-2** |
| **Total Credit** | **04 (03 Theory + 01 Practical)** |
| **Total Marks** | **100(75Theory + 25 Practical)** |
| **Marks distribution of the Theory paper** | **End semester = 67**<br>**Internal Marks = 08** |
| **Marks distribution of the Practical paper** | **End semester = 22**<br>**Internal Marks = 03** |
| **Contact Hours** | **Theory: 3 Hrs. per week**<br>**Practical: 2 Hrs. per week** |

**Applicable to the students**

1. **Those who have opted for chemistry as a Major subject in Table 1 or Table 2**
2. **Those who have opted for chemistry asaMinor subject in Table 1 or Table 2**

**LEARNING OUTCOMES SPECIFIC TO DISCIPLINE**

By the end of the course, students will be able to:

- Explain various theories in covalent-bonded simple inorganic compounds.
- Explain the different concepts regarding structures in ionic solids, their rules and limitations. Lattice energy, Born-Haber cycle, Fajan's rule and Van der Waals forces.
- Understand and explain various methods for preparation, chemical reactions of alkanes and cycloalkanes. Concept of reactivity/selectivity of alkane halogenation & strain theories of cycloalkanes.
- Explain the factors that affect optical isomerism in organic compounds.
- Know the fundamentals of the kinetic theory of gases, Maxwell-Boltzmann distribution law, mean free path of molecular collisions, viscosity, thermal conductivity and diffusion.
- Demonstrate gas laws and isotherms of Van der Waal's, get an essence of the basic concepts of thermodynamics, the first law of thermodynamics, the concept of entropy and the associated theorems, the thermodynamic potentials and their physical interpretations along with Joule-Thomson coefficient and inversion temperature.
- Experimentally the students will be able to explain the concept of standardisation of primary standards, determination of strength & water of crystallisation.

**GENERIC OUTCOMES**

The first year of the course is designed to inculcate a scientific temper amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out analysis, evaluation and draw logical conclusions.To hone their critical thinking and enhance creativity, which shall lead to innovative and out-of-the-box thinking, which shall help in complex problem-solving situations in higher classes.They are also encouraged to work and improve upon their communication skills, through proper listening, literature survey and expression through word and print.

**Semester-2**
**Syllabus for Theory Paper**
**CHM-DSC-2-MAJ/MIN-201**
**General Chemistry-2**

| | |
|---|---|
| **End semester = 67** | **Credits - 03** |
| **Internal Marks = 08** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

## UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

### Chemical Bonding – I (8 Hrs.)

Covalent Bond – Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridisation and shapes of simple inorganic molecules and covalent ions. $BeF_2$, $BF_3$, $CH_4$, $PF_5$, $SF_6$, $IF_7$, $SnCl_2$, $XeF_4$, valence shell electron pair repulsion (VSEPR) theory to $H_3O^+$, $SF_4$, $ClF_3$, $ICl^-$ and $H_2O$. MO theory, homonuclear (elements and ions of $1^{st}$and $2^{nd}$ row), and heteronuclear(CN, $CO^+$, CO, $CN^-$), diatomic molecules, Percentage ionic character from dipole moment and electronegativity.

### Chemical Bonding – II (7 Hrs.)

Ionic Solids – Concept of close packing, Ionic structures (NaCl type, and Zinc blende),radius ratio rule and coordination number, limitation of radius ratio rule, lattice defects, semiconductors.

Lattice energy and Born-Haber cycle, solvation energy and solubility of ionic solids, polarising power and polarisability of ions, Fajan'srule.Metallic bond-free electron, valence bond and band theories.WeakInteractions – Hydrogen bonding, Van der Waals forces.

## UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

### Alkanes and Cycloalkanes (8 Hrs.)

Structure, IUPAC Nomenclature and Isomerism, Methods of Preparation (Sabatier and Senderen's reaction, from alkyl halide, Wurtz reaction, Corey House synthesis, Kolbe's method, decarboxylation), Physical and Chemical Properties, Mechanism of Halogenation of Alkanes-Relative reactivities of halogens, alkanes, and different classes of Hydrogen atom; Orientation, Reactivity and Selectivity.

Cycloalkanes, nomenclature, methods of formation, chemical reactions, Baeyer's strain theory and its limitation, ring strain in small rings (cyclopropane and cyclobutane), theory of strainless rings, the case of cyclopropane ring: banana bonds.

### Stereochemistry and Optical Isomerism (7 Hrs.)

Concept of Isomerism, Types of isomerism,Tetrahedral concept of carbon, Polarimetry, specific rotation. Optical isomerism- Elements of Symmetry, Molecular Chirality, Enantiomerism, stereogeniccentre, optical activity, chiral and achiral molecules with two stereogeniccentres, Diastereoisomers, Erythro and Threo isomers, Meso compounds. Representation of Stereoisomers-Saw-horse, Newman, Flying edge and Fischer Projection. Number of stereoisomers in chiral compound, Racemisation, Internal and External compensation, Resolution of racemic mixture, Enantioselective synthesis, Walden inversion, Relative and absolute configuration, sequence rules, D & L and R & S system of nomenclature (with examples)

## UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

### Gaseous States (8 Hrs.)

Gas laws, Idea and real gases, postulates of kinetic theory of gases, deviation from ideal behavior, Van der Waal's equation of state, Critical Phenomena:PV isotherms of real gases, the isotherms of Van der Waal's equation, relationship between critical constants and Van der Waal's constants, the law of corresponding states, reduced equation of state. MolecularVelocities: Root mean square, average and most probable velocities. Collision number, mean free path and collision diameter, Liquification of gases (based on Joule-Thomson effect).

### Thermodynamics-I (7 Hrs.)

Definition of Thermodynamic Terms: System, surroundings, etc. Types of systems, intensive and extensive properties, State and path functions and their differentials.Thermodynamic process.Concept of heat and work. First Law of Thermodynamics: Statement, definition of internal energy and enthalpy, Heat capacity, heat capacities at constant volume and pressure and their relationship, Joule's Law-Joule-Thomson coefficient and inversion temperature. Calculations of w, q, dU and dH for the expansion of ideal gases under isothermal and adiabatic conditions for a reversible process.

### Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd.,1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Mileston es Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson education, 2008.

20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

## Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

## Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

**Semester-2**
**Syllabus for Practical Paper**
**CHM-DSC-2-MAJ/MIN-201**
**General Chemistry-2**

| | |
|---|---|
| **End semester = 22** | **Credit - 01** |
| **Internal Marks = 03** | **Contact Hours – 2/week** |
| **Examination Time allowed** | **3 Hours** |

## I. INORGANIC CHEMISTRY

1. Permanganometry and dichromatometry
    a. Standardization of $KMnO_4$ solution
    b. Determination of the strength of potassium oxalate-oxalic acid mixture/Mohr's salt/ferrous sulphatesolution by usingastandard solution of $KMnO_4$ solution by redox titration.
    c. Determination of water of crystallisation in Mohr's salt and ferrous sulphatebyusingastandard solution of $KMnO_4$ by redox titration.

## II. ORGANIC CHEMISTRY

1. Concept of induction of crystallisation.
2. Phthalic acid from hot water (using fluted filter paper and a stemlessfunnel).
3. Acetanilide from boiling water.
4. Benzoic acid from water

## III. PHYSICAL CHEMISTRY

Surface Tension: To determine the surface tension of Brine Solution (20%), n-Butyl alcohol, cyclohexane

### Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

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# Computer Science and Applications

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY(Table 2*)**

**BSc Computer Science subjects listed for first four semesters till date**

**Table 1***

| Sem | Level | Course Code | Course Name | Credits (Th+Pr) | | Total Credits |
|---|---|---|---|---|---|---|
| I | L-100 | CSA-DSC-1(Maj/Min)-101 | Computer Fundamentals and PC Software | 2+2 | 4(x3) | 24 |
| | | CSA-MDC-101/201/301 | E-Commerce | 3+0 | 3 | |
| | | CSA-MDC-102/202/302 | Management Information System | 3+0 | | |
| | | CSA-MDC-103/203/303 | Web Development/ Designing | 0+3 | | |
| | | CSA-MDC-104/204/304 | Introduction to Information Technology | 0+3 | | |
| | | CSA-VAC-101/201/301/401 | Introduction to Blockchain Technology | 2+0 | 2 | |
| | | CSA-VAC-102/202/302/402 | Introduction to Computer Hardware | 2+0 | | |
| | | CSA-VAC-103/203/303/403 | Digital Empowerment | 2+0 | | |
| | | CSA-VAC-104/204/303/403. | Ethics of Artificial Intelligence | 2+0 | | |
| | | CSA-SEC-101/201/301 | Computer Fundamentals and Personal Computing Software | 0+3 | 3 | |
| | | CSA-SEC-102/202/302 | Desktop Publishing using GIMP | 0+3 | | |
| | | CSA-SEC-103/203/303 | Backend Web Development | 0+3 | | |
| | | CSA-SEC-104/204/304 | Business Intelligence and Data Visualization | 0+3 | | |
| | | CSA-SEC-105/205/305 | Digital Marketing | 0+3 | | |
| | | CSA-SEC-106/206/306 | Computer Literacy | 0+3 | | |
| | | CSA-SEC-107/207/307 | Desktop Publishing Using Photoshop | 0+3 | | |
| | | CSA-SEC-108/208/308 | Fundamentals of Web Designing | 0+3 | | |
| | | CSA-SEC-109/209/309 | Application Software for Office Automation | 0+3 | | |
| | | CSA-SEC-110/210/310 | Python Programming | 0+3 | | |
| | | (AEC-1-101) | Eng-1 | 2 | | |
| | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | | |

**Table 2***

| Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | | Total Credits |
|---|---|---|---|---|---|---|
| I | L-100 | CSA-DSC-1(Maj/Min)-101 | Computer Fundamentals and PC Software | 2+2 | 4(x3) | 24 |
| | | CSA-MDC-101/201/301 | E-Commerce | 3+0 | 3 | |
| | | CSA-MDC-102/202/302 | Management Information System | 3+0 | | |
| | | CSA-MDC-103/203/303 | Web Development/ Designing | 0+3 | | |
| | | CSA-MDC-104/204/304 | Introduction to Information Technology | 0+3 | | |
| | | CSA-VAC-101/201/301/401 | Introduction to Blockchain Technology | 2+0 | 2 | |
| | | CSA-VAC-102/202/302/402 | Introduction to Computer Hardware | 2+0 | | |
| | | CSA-VAC-103/203/303/403 | Digital Empowerment | 2+0 | | |
| | | CSA-VAC-104/204/304/404 | Ethics of Artificial Intelligence | 2+0 | | |
| | | CSA-SEC-101/201/301 | Computer Fundamentals and Personal Computing Software | 0+3 | 3 | |
| | | CSA-SEC-102/202/302 | Desktop Publishing using GIMP | 0+3 | | |
| | | CSA-SEC-103/203/303 | Backend Web Development | 0+3 | | |
| | | CSA-SEC-104/204/304 | Business Intelligence and Data Visualization | 0+3 | | |
| | | CSA-SEC-105/205/305 | Digital Marketing | 0+3 | | |
| | | CSA-SEC-106/206/306 | Computer Literacy | 0+3 | | |
| | | CSA-SEC-107/207/307 | Desktop Publishing Using Photoshop | 0+3 | | |
| | | CSA-SEC-108/208/308 | Fundamentals of Web Designing | 0+3 | | |
| | | CSA-SEC-109/209/309 | Application Software for Office Automation | 0+3 | | |
| | | CSA-SEC-110/210/310 | Python Programming | 0+3 | | |
| | | (AEC-1-101) | Eng-1 | 2 | | |
| | | (PUN-COMPL-1-101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| II | L-100 | CSA-DSC-2(Maj/Min)-201 | Programming in C | 2+2 | 4 (x3) | 24 | II | L-100 | CSA-DSC-2(Maj/Min)-201 | Programming in C | 2+2 | 4 (x3) | 24 |
| | | CSA-MDC-101/201/301 | E-Commerce | 3+0 | 3 | | | | CSA-MDC-101/201/301 | E-Commerce | 3+0 | 3 | |
| | | CSA-MDC-102/202/302 | Management Information System | 3+0 | | | | | CSA-MDC-102/202/302 | Management Information System | 3+0 | | |
| | | CSA-MDC-103/203/303 | Web Development/ Designing | 0+3 | | | | | CSA-MDC-103/203/303 | Web Development/ Designing | 0+3 | | |
| | | CSA-MDC-104/204/304 | Introduction to Information Technology | 0+3 | | | | | CSA-MDC-104/204/304 | Introduction to Information Technology | 0+3 | | |
| | | CSA-VAC-101/201/301/401 | Introduction to Blockchain Technology | 2+0 | 2 | | | | CSA-VAC-101/201/301/401 | Introduction to Blockchain Technology | 2+0 | 2 | |
| | | CSA-VAC-102/202/302/402 | Introduction to Computer Hardware | 2+0 | | | | | CSA-VAC-102/202/302/402 | Introduction to Computer Hardware | 2+0 | | |
| | | CSA-VAC-103/203/303/403 | Digital Empowerment | 2+0 | | | | | CSA-VAC-103/203/303/403 | Digital Empowerment | 2+0 | | |
| | | CSA-VAC-104/204/304/404 | Ethics of Artificial Intelligence | 2+0 | | | | | CSA-VAC-104/204/304/404 | Ethics of Artificial Intelligence | 2+0 | | |
| | | CSA-SEC-101/201/301 | Computer Fundamentals and Personal Computing Software | 0+3 | 3 | | | | CSA-SEC-101/201/301 | Computer Fundamentals and Personal Computing Software | 0+3 | 3 | |
| | | CSA-SEC-102/202/302 | Desktop Publishing using GIMP | 0+3 | | | | | CSA-SEC-102/202/302 | Desktop Publishing using GIMP | 0+3 | | |
| | | CSA-SEC-103/203/303 | Backend Web Development | 0+3 | | | | | CSA-SEC-103/203/303 | Backend Web Development | 0+3 | | |
| | | CSA-SEC-104/204/304 | Business Intelligence and Data Visualization | 0+3 | | | | | CSA-SEC-104/204/304 | Business Intelligence and Data Visualization | 0+3 | | |
| | | CSA-SEC-105/205/305 | Digital Marketing | 0+3 | | | | | CSA-SEC-105/205/305 | Digital Marketing | 0+3 | | |
| | | CSA-SEC-106/206/306 | Computer Literacy | 0+3 | | | | | CSA-SEC-106/206/306 | Computer Literacy | 0+3 | | |
| | | CSA-SEC-107/207/307 | Desktop Publishing Using Photoshop | 0+3 | | | | | CSA-SEC-107/207/307 | Desktop Publishing Using Photoshop | 0+3 | | |
| | | CSA-SEC-108/208/308 | Fundamentals of Web Designing | 0+3 | | | | | CSA-SEC-108/208/308 | Fundamentals of Web Designing | 0+3 | | |
| | | CSA-SEC-109/209/309 | Application Software for Office Automation | 0+3 | | | | | CSA-SEC-109/209/309 | Application Software for Office Automation | 0+3 | | |
| | | CSA-SEC-110/210/310 | Python Programming | 0+3 | | | | | CSA-SEC-110/210/310 | Python Programming | 0+3 | | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | | (AEC-2-201) | Eng-2 | 2 | | |
| | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | | | | | (PUN-COMPL-2-201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II | | | | | | | | | | | | | |
| The student exiting the programme after one year, securing48credits,will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. | | | | | | | | | | | | | |

***Note: Student need to select one MDC, VAC, SEC out of given options, respectively.**
**NO MDC, SEC, VAC will be repeated. All MDC, SEC and VAC are Level-100**

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3$^{rd}$ semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

# Paper Title : Computer Fundamentals and PC Software
# Paper Code: CSA-DSC-1(Maj/Min)-101

| Th | Pr | Cr |
|---|---|---|
| 2 | 2 | 4 |

**Theory External Marks: 45**
**Theory Internal Marks: 05**
**Number of Th. Hrs.: 30**
**Practical Ext. Marks: 50**
**Number of Pr. Hrs.: 60**

**Time Duration: 3 Hrs.**
**Level: L-100**

**Objectives:** The objective of this course is to familiarize students with the fundamentals of computers and packages commonly used in application software.

**Learning Outcome:**

On successful completion of the course, students will be able to:

i. Understand the fundamentals of personal computers, application usages etc.
ii. Get the knowledge of peripherals of personal computer and booting process.
iii. Use word processing packages for writing emails, letters and chapters etc.
iv. Handle statistical data using spreadsheets packages.
v. Design presentations for business analyses, industrial growth etc.

**Note :**

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

## UNIT - I

**Introduction to Computers:** Characteristics of Computers, Functional Units of Computer System: CPU, registers, system bus, main memory unit, cache memory; Inside a computer: SMPS, Motherboard, Ports and Interfaces, expansion cards, ribbon cables, memory chips, processors

**Devices:** Input and output devices, Keyboard, Mouse, Joystick, Scanner, OCR, OMR, Bar Code Reader, Web Camera, Monitor, Printer and Plotter; Memory: Primary, Secondary/Auxiliary memory, RAM and ROM and their types, Cache memory, Units of measurement of storage; Storage devices, Hard disks, Optical disks, Memory storage devices: Flash Drive, Memory card; Types of software: System and Application software

**Data Representation:** Bit, Byte, Binary, Decimal, Hexadecimal, and Octal Systems. Conversions between different number systems; Character Codes: Introduction to ASCII, EBCDIC and Unicode character sets

**Understanding Windows Operating System:** Anatomy of windows, Taskbar Settings, managing folders and files, Searching Files and folders, Customizing Windows, Recycle Bin, Control Panel.

## UNIT - II

**Introduction to Word Processing:** Opening, Creating, Saving, Printing and closing Documents, Using the Interface (Menu Toolbars), Editing Text (Copy, Delete, Move), Finding and Replacing Text, Spell Check, Autocorrect; Auto-text, Character formatting, Paragraph formatting, Page formatting.

**Document Enhancement:** Inserting Pictures, Tables, inserting special characters, Adding Borders and shading, Adding Headers and Footers, numbering pages, setting up Multiple columns, Adjusting Margins and Hyphenating Documents, changing document views, Adding comments to a document, Tracking changes to a document.

**Using styles and templates:** Applying, modifying and creating new (custom) styles; Creating a table of contents, creating indexes and bibliographies

**Mail Merge and Macros:** Creating Master Document and Data Source, Merging and printing Documents; Creating and using Macros, Creating fill-in forms.

**UNIT - III**

**Introduction to Spreadsheets:** Worksheet overview, Row, Column, Cells, Ribbon, Tabs, Groups, Name Box and Formula Bar; Creating, Opening, Saving and printing worksheet; Speeding up data entry: using fill tool, fill series, defining fill series; working with Formulae, Data formatting (number formatting, date formatting),Working with Ranges, Absolute, relative and Mixed addressing; Viewing a spreadsheet: freezing rows and columns, splitting screen; creating, sorting and filtering Data Base; Creating and running Macros, Assigning shortcuts to Macros

**Using Functions:** Statistical, financial, Mathematical, string, date and time.

**Working with sheets:** inserting new sheet, moving, deleting and renaming sheets

**Creating Charts:** Understanding Chart types, Creating Embedded Charts, formatting Charts: Changing Chart types, adding Titles, Legends and Gridlines, Printing Charts

**UNIT – IV**

**Presentation Software:** Creating, saving, and printing presentations; selecting design templates, Changing Background and Layout, Inserting tables and images, animations and transitions

**Databases:** Introduction to database, Creating database using Wizard or from scratch, creating tables using wizard, entering data, using design view, saving, inserting, editing, Changing properties of fields, setting primary key.

**Suggested Readings:**

1. Sinha P.K. & Sinha Priti, Computer Fundamentals, BPB Publications
2. Prof. Satish Jain, Shashi Singh, M. Geetha, Windows 10 with MS office 2016, BPB Publications
3. Mansfield Ron,Working in MS Office, Tata McGraw Hill.

## Title : Practical Based on Paper – CSA-DSC-1(Maj/Min)-101 (Computer Fundamentals and PC Software)

## Paper Code: CSA-DSC-1(Maj/Min)-101P

**Time Duration: 3 Hrs.** **Number of Hours: 60**

**Note :**Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

# Paper Title: Programming in C
# Paper Code :CSA-DSC-2(Maj/Min)-201

| Th | Pr | Cr |
|---|---|---|
| 2 | 2 | 4 |

**Theory External Marks: 45**
**Theory Internal Marks : 05**
**Number of Th. Hrs.: 30**
**Practical Ext. Marks: 50**
**Number of Pr. Hrs.: 60**

**Time Duration: 3 Hrs.**
**Level: L-100**

**Objectives:** The objective of this course is to familiarize students with programming language ”C” and its use in Problem solving.

**Learning Outcome:**
On successful completion of the course, students will be able to:

i. Understand the programming concepts.
ii. Learn the C programming in practical usage.
iii. Solve programming problems using C programming language.
iv. Gain the in depth knowledge of strings, arrays, functions and file handling etc.
v. Develop applications based on programming concepts.

**Note :**

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

## UNIT - I

**Programming Process:** Steps in developing of a program, Algorithm development, Flowchart, Testing and Debugging.
**Fundamentals of C Languages:** History of C, Character Set, Identifiers and Keywords, Constants, Types of C Constants, Rules for Constructing Integer, Real and character Constants, Variables, Data Types, rules for constructing variables.

**Operators and Expressions:** C Instructions, Arithmetic operators, Relational operators, Logical operators, Assignment Operators, Type Conversion in Assignments, Hierarchy of Operations, Standard and Formatted Statements, Structure of a C program, Compilation and Execution.

## UNIT - II

**Decision Control Structure:** Decision making with IF-statement, IF-Else and Nested IF-Else, The else if Clause. Loop Control Structure: While and do-while, for loop and Nested for loop, Case Control Structure: Decision using switch, The goto statement.
**Functions:** Library functions and user defined functions, Global and Local variables, Function Declaration, Calling and definition of function, Methods of parameter passing to functions, recursion.

## UNIT - III

**Arrays:** Introduction, Array declaration, accessing values in an array, initializing values in an array, Single- and Two-Dimensional Arrays, initializing a 2-Dimensional Array, Memory Map of a 2-Dimensional Array, Arrays of characters, Insertion and deletion operations, Searching the elements in an array, Using matrices in arrays.
**Pointers:** Pointer declaration, Address operator “&”, Indirection operator “*”, Array of Pointers.

**UNIT - IV**

**Storage Classes in C :** Auto, Extern ,Register and Static with their Scope, Storage & Lifetime.
**String Manipulation in C:** Declaring and Initializing string variables, Reading and writing strings, String Handling functions (strlen(), strcpy(), strcmp(), strcat()).

**Structures and Unions:** Definition. Advantages of Structure, Declaration of structures, Structure Initialization, Accessing Structure members, Arrays of Structures, Union Definition, Difference between Structure and Union.

**Suggested Readings:**

1. Yashavant P. Kanetkar : Let us C, BPB Publications, New Delhi.
2. Salaria, R.S. : Test Your Skills in C, Salaria Publications, New Delhi.
3. C. Balaguruswami : Programming with C Language, Tata McGraw Hill, New Delhi.
4. Byron S. Gottfried : Programming in C, McGraw Hills Publishers, New York.
5. M.T. Somashekara : Programming in C, Prentice Hall of India.

## Title : Practical Based on Paper – CSA-DSC-2(Maj/Min)-201 (Programming in C)

## Paper Code: CSA-DSC-2(Maj/Min)-201P

**Time Duration: 3 Hrs.** **Number of Hours: 60**

**Note :**Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

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# ENVIRONMENT CONSERVATION

## SEMESTER-I
## MINOR COURSE
## ENVIRONMENTAL CONSERVATION: NATURAL RESOURCE MANAGEMENT

**Course Code: MINC1** **Credits= 4 (T=3 + P=1)**

**Course Title: Environmental Conservation: Natural Resources management**
**Total Hours = 60 (Theory=45+Practical=15)**
**Total Marks = 100 (Theory=75+Practical=25)**
**Internal Assessment =20%**

***Course Outcomes***

1. *The students will be able to understand about natural resources, their types and realize the importance of resources.*
2. *The students will be able to acquire deep knowledge about land and forest resources.*
3. *The students will be able to describe energy resources.*
4. *The students will be able to analyse Indian as well as global statusof natural resources.*
5. *The students will be able to study conservation methods of natural resources.*

## THEORY

**Unit-I (15 Hrs)**
**Natural and Energy Resources:**
1.1 Natural resources-Classification and types of resources
1.2 Renewable and non-renewable resources
1.3 Concept of conventional and non-conventional sources of energy
1.4 Fossil fuels- coal, crude oils and natural gas
1.5 Solar energy
1.6 Wind energy
1.7 Biomass energy
1.8 Geothermal energy

**Unit-II (15 Hrs)**
**Forest Resources**
2.1 Importance of forest;
2.2 Productive functions, protective and ameliorative functions,
2.3 Non-timber forest products
2.4 Major forest types in India
2.5 Conservation of forest resources
2.6 Global status of forest conservation efforts

**Unit-III (15 Hrs)**
**Land Resources**
3.1 Land Resources: Global land use patterns
3.2 Soil degradation – causes; Soil erosion, Desertification
3.3 Wasteland Reclamation. Land degradation
3.4 Land cover, land use changes; land degradation and its control
3.5 Water logging and salinity
3.6 Organic farming

## PRACTICALS (15 Hrs)

1. Prepare a report on promotion of green energy in India
2. To do a survey of water consumption and conservation practises of a selected area.
3. To visit and make a list of forest and non-forest products in the market.
4. Identify the sources of energy used in your area
5. Visit to a local area to document environmental assets river/ forest/grassland/hill/mountain
6. To determine soil quality parameters using portable soil analysis kit

## SUGGESTED READINGS:

1. Rogoff, M., Screve, F. 2019. Waste-to-Energy, Elsevier, Amsterdam
2. Singh, M.P, Rallan,B.R. and Vivek Kumar. 2012. Natural Resources Management. Emkay Publishing House.
3. Odum, E.P. and Barrett, G.W. 2008. Fundamentals of Ecology (5thEdition). Thomson Brooks Australia and Affiliated to East West Press Pvt. Ltd., New Delhi.
4. Kanagasabai, T. 2010. Environmental Studies. PHI Learning Private Limited, New Delhi. 5. Sharma, P.D. 2018. Ecology and Environment. Rastogi Publication, Meerut.
6. Joseph, K. and Nagendran, R. 2004. Essentials of Environmental Studies. Pearson Education, New Delhi.

## SEMESTER-II
## MINOR COURSE
## ENVIRONMENTAL CONSERVATION: ENVIRONMENTAL POLLUTION AND MANAGEMENT

**CourseCode:MINC2** **Credits= 4 (T=3 + P=1)**

**CourseTitle: Environmental Conservation: Environmental Pollution and Management**
**TotalHours=60(Theory=45+Practical=15)**
**TotalMarks=100(Theory=75+Practical=25)**
**InternalAssessment=20%**

***Course Outcomes***

1. *The students will be able to understand about causes, sources and effects of environmental pollution.*
2. *Thestudentswillbeableto learn about atmospheric structure and processes*
3. *ThestudentswillacquireknowledgeaboutGlobal Water Balance and Hydrological cycle.*
4. *Thestudentswill get practical knowledge of water quality parameters and soil parameters.*

## THEORY

**Unit-I(15 Hrs)**
**Hydrosphere and Water Resources**
1.1 Composition of Hydrosphere, Global Water Balance
1.2 Hydrological Cycle; Major Sources of Water, Surface and subsurface water sources
1.3 Water Anomalies including Floods and Droughts
1.4 Overutilization of Water and Water Scarcity
1.5 Conflicts Over Water – National and International
1.6 Case Studies; Satluj Yamuna Link Canal Issue, Kaveri Issue

**Unit-II (15 Hrs)**
**Water Pollution and Control Measures**
2.1 Water Pollution: Concept, Causes and Sources
2.2 Water Pollutants and categories
2.3 Effect of water pollution on plants , animals and humans
2.4 Basics of Water Testing , Potable water
2.5 Control Measures; Waste Water Treatment
2.6 Root Zone technology

**Unit-III(15 Hrs)**
**Air Pollution and Control Measures**
3.1 Composition and Structure of Atmosphere
3.2 Air Pollution: Concept and Sources; Noise Pollution
3.3 Air Pollutants and Categories
3.4 Effects of Air Pollution
3.5 Major Air Pollution Issues: Global Warming, Acid rain, Ozone Depletion
3.7 Control Measures; Air Pollution Control Devices

### PRACTICALS (15 Hrs)

1. Draw the pie diagram showing the global water balance.
2. To study the structure of the atmosphere with the help of a label diagram
3. To determine water quality parameters (pH, EC and TDS) using analysis kit.
4. To study air quality parameter like SPM using Air Quality Monitor
5. To visit, collect data and prepare report regarding air and water pollution of a site.

## SUGGESTED READINGS:

1. Joseph, K. and Nagendran, R.(2004). Essentials of Environmental Studies. Pearson Education, New Delhi.
2. Cunningham, W.P. and Cunningham, M.A. (2002*). Environmental Science: Inquiry andApplications.* AGlobalConcern.TataMcGraw-HillPublishingCompany,NewDelhi.
3. De. AK, (2003). Environmental Chemistry. New Age International, New Delhi
4. Khitoliya, RK (2012) Textbook on Environmental Pollution. S Chand Publishing, New Delhi
5. American Public Health Association, Standard Methods for Examination of Water and Wastewater, APHA, New York, 2003.
6. Sengupta, B., John, S. & Abraham, T. Forth Estate, A Handbook on Air, Water and Land Pollution. IDMA Foundation for Sustainable Development and BR Publications Corporation, Delhi. 2006.
7. Aggarwal, A. K., Noise Pollution. APH Publications, New Delhi, 2008.
8. Dara, S.S., A Textbook of Environmental Chemistry. 2nd Edition. S. Chand & Company Ltd., Ram Nagar, New Delhi, 2010.
9. Hill, M. K., Understanding Environmental Pollution, Cambridge Univ. Press, 2008.
10. Aggarwal, A. K., Water Pollution, APH Publications, New Delhi, 2008

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# Industrial Chemistry

| CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MAJOR** SUBJECT AS PER TABLE 2 | | | | | |
|---|---|---|---|---|---|
| **Table 2** | | | | | |
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** | **Total Credits** |
| I | L-100 | (ICH-DSC-1 MAJ/MIN-101) | Basic Industrial Principles, Data Analysis and Laboratory Operations | 4 | 24 |
| | | B1 | To be chosen by the student | 4 | |
| | | C1 | To be chosen by the student | 4 | |
| | | IDC-1 | To be chosen from the available options | 3 | |
| | | SEC-1 | To be chosen from available options | 3 | |
| | | ENG-1 | Compulsory | 2 | |
| | | VAC-1 | To be chosen from the common basket | 2 | |
| | | PBI-1/HCP-1 | Compulsory | 2 | |
| II | L-100 | (ICH-DSC-2 MAJ/MIN-201) | Applied Industrial Chemistry | 4 | 24 |
| | | B2 | To be chosen by the student | 4 | |
| | | C2 | To be chosen by the student | 4 | |
| | | IDC-2 | To be chosen from the available options | 3 | |
| | | SEC-2 | To be chosen from available options | 3 | |
| | | ENG-2 | Compulsory | 2 | |
| | | VAC-2 | To be chosen from the common basket | 2 | |
| | | PBI-2/HCP-2 | Compulsory | 2 | |

| CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MINOR**SUBJECT AS PER TABLE 1 AND TABLE 2 | | | | |
|---|---|---|---|---|
| **Table 2** | | | | |
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits** |
| I | L-100 | (ICH-DSC-1 MIN/MAJ-101) | Basic Industrial Principles, Data Analysis and Laboratory Operations | 4 |
| II | L-100 | (ICH-DSC-2 MIN/MAJ-201) | Applied Industrial Chemistry | 4 |

**Note:-**

1. In the first semester, students are offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3$^{rd}$ semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

| | |
|---|---|
| **Semester** | **I** |
| **Code of Paper** | **ICH-DSC-1-MAJ/MIN-101** |
| **Title of Paper** | **Basic Industrial Principles, Data Analysis and Laboratory Operations** |
| **Total Credit** | **04 (03 Theory + 01 Practical)** |
| **Total Marks** | **100 (75 Theory + 25 Practical)** |
| **Marks distribution of the Theory paper** | **End semester = 67**<br>**Internal Marks = 08** |
| **Marks distribution of the Practical paper** | **End semester = 22**<br>**Internal Marks = 03** |
| **Contact Hours** | **Theory: 3 Hrs. per week**<br>**Practical: 3 Hrs. per week** |

**Applicable to the students**

3. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
4. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

**LEARNING OUTCOMES SPECIFIC TO DISCIPLINE**

By studying this course, the students will be able to:

- Explain the what are the basic dimensions and units.
- The student shall be able to carry out the important chemical calculations and determine the errors that may result in the data.
- Explain the theory behind chemical concentrations, preparation of standards and analyse data.
- The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

**GENERIC OUTCOMES**

The first year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

**For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.**

# Semester-1
# Syllabus for Theory Paper
## ICH-DSC-1-MAJ/MIN-101

### Basic Industrial Principles, Data Analysis and Laboratory Operations

| | |
|---|---|
| **End semester = 67** | **Credits - 03** |
| **Internal Marks = 08** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

### UNIT-1 (15 Hrs.)

**Dimensions and Units -** Introduction, Definitions of the Basic Units - (Mass, Length, Time, Temperature, Volume, Pressure), Basic terminologies, Characteristics of standard units, Classification of units: Fundamental and Derived units, Systems of units, Basic and supplementary units, Conversion between units and conversion factors, Metric prefixes. Definition of Dimensions, Dimensional Formula, Dimensional Equation, Dimensional Constant, Dimensional Variable, Dimensionless Constant, Dimensionless Variable, Primary Dimensions and units, Secondary Dimensions and units, Dimensional Analysis, Dimensional Homogeneity, Applications of Dimensional Equation, Limitations of Dimensional Equation.

**Basic Chemical Calculations** - Atomic weight, Molecular weight, Equivalent weight, Mole, Mole fraction, Mole percent, Mass fraction of solution, Weight percentage, Density, Concentration, Specific gravity, Volumetric flow rate, Analyses of Multi Components Solutions and Mixtures.

**Errors in Chemical Analysis -** Definition of errors, mean and median, Accuracy and Precision in experimental data, Types of errors: Absolute and relative error, Analytical errors: Random, systematic and gross errors, Sources and factors affecting, detection and elimination of errors, significant figures.

### UNIT-2 (15 Hrs.)

**Chemical Concentration –** Definition, Concentration Units: Percentage (mass percent, volume percent, Mass-volume percent), Parts per millions (ppm), Strength of solution, Molarity, Molality, Normality, Formality, Factors affecting concentration, Concentration unit conversion and problem-solving in chemical engineering involves conversion between different concentration units like molarity, molality, normality, parts per million (ppm), and mass/volume percentages, Formula weight and molecular weight, Molecular formula and Empirical formulas.

**Preparation of solutions -**Basic Concepts of preparing solutions, Standard solutions, Primary standards, Secondary standards, Buffer Solutions, Concentrated solutions, Preparation of simple inorganic salt solutions, Preparation of acid solutions, Preparation of Base Solutions.

**Stoichiometric Analysis** – Calculation of the quantitative relationships between reactants and products in chemical reactions, Reaction stoichiometry, Balanced Chemical Equations for complete and partial combustion of carbon and hydrocarbons, Stoichiometric Coefficients, Stoichiometric Ratios, concept of limiting reactant and excess reactants in a reaction mixture with examples, Yield.

**Data analysis and Chemical drawing and visualization tool** - To plot linear and non-linear equations using Excel spreadsheet, Analyzing and interpreting experimental data using origin software, Calibration curve plotting using excel and origin software, draw and visualize chemical structures using ChemDraw and ChemSketch.

## UNIT-3 (15 Hrs.)

**Common Laboratory Apparatus and Calibration** – Theory of Analytical Balance, Operation and theory of the balance, Construction details, Components, Sources of Errors and Error Analysis, Maintenance Guidelines, identification, proper usage, and calibration techniques of various laboratory tools and equipment.

**pH meter** - Introduction to pH and pH Measurement, Principle of pH Measurement, pH electrode components, use of pH Meter, Calibration using buffer solutions, Troubleshooting and maintenance of pH meter, application of data Factors affecting pH measurement.

**Fundamentals of Laboratory Apparatus -** Types of Laboratory apparatus: Volumetric flasks, burettes, pipettes, weighing bottles, specific gravity bottle, different types of funnels, chromatographic columns, chromatographic jars, desiccators, drying ovens, filter crucibles, rubber policeman, use of volumetric glass ware, Importance of laboratory apparatus.

**Safety with chemicals and waste** - Personal Protective Equipment (PPE), Handling hazardous materials, Emergency Procedures.

### Books Recommended

1. Higson, S. P.J. (2003), Analytical Chemistry, Oxford University Press.
2. Skoog, D.A., West, D.M. (2003), Fundamentals of Analytical Chemistry, Brooks/Cole.
3. Christian, G.D. (2004), Analytical Chemistry, 6th Edition, John Wiley & Sons, New York.
4. Fifield, F.W., Kealey, D. (2000), Principles and Practice of Analytical Chemistry, Wiley.
5. Dean, J. A. (1997), Analytical Chemistry Handbook, McGraw Hill.
6. Day, R. A., Underwood, A. L. (1991), Quantitative Analysis, Prentice Hall of India.
7. Gordus, A. A. (1985), Schaum's Outline of Analytical Chemistry, Tala McGraw-Hill.
8. Harris, D. C. (2007), Exploring Chemical Analysis, W.H. Freeman and Co.

### Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

### Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

# Semester-1
# Syllabus for Practical Paper
## ICH-DSC-1-MAJ/MIN-101

### Basic Industrial Principles, Data Analysis and Laboratory Operations

| | |
|---|---|
| **End semester = 22** | **Credit - 01** |
| **Internal Marks = 03** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

**Basic Industrial Laboratory Operations:**

1. Use and calibration of volumetric equipment and glassware.
2. Preparation of standard solutions of NaOH, $KMnO_4$, $H_2SO_4$, HCl, $Na_2CO_3$, Oxalic acid. Introduction to primary and secondary standards.
3. Preparation of pure sample of potash alum.

**Data Analysis and Laboratory Operations:**

4. To plot accuracy graphfor each value of the independent variable using:
   (a) Excel data sheet (b) Origin software
5. To determine the errors in experimental data by plotting error bar graphs using:
   (a) Excel data sheet (b) Origin software
6. Calibration of equipment.
7. Preparation of buffer solutions.
8. Use of pH meter: determination of pH of given solutions.
9. Power point presentations.

**Books Recommended**

1. Seamus P.J. Higson: Analytical Chemistry.
2. Douglas A. Skoog and Donald M.West: Fundamentals of Analytical Chemistry.
3. Adion A. Gordus: Schaum's Outline of Analytical Chemistry, Tata McGraw-Hill.
4. Gary D. Christian: Analytical Chemistry.
5. Freifelder and Kealy: Analytical Chemistry.
6. Daniel C Harris: Exploring Chemical Analysis.
7. Daniel C Harris: Quantitative Chemical Analysis.

**General Instructions to the Examiners:**

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

| | |
|---|---|
| **Semester** | **II** |
| **Code of Paper** | **ICH-DSC-2-MAJ/MIN-201** |
| **Title of Paper** | **Applied Industrial Chemistry** |
| **Total Credit** | **04 (03 Theory + 01 Practical)** |
| **Total Marks** | **100(75Theory + 25 Practical)** |
| **Marks distribution of the Theory paper** | **End semester = 67**<br>**Internal Marks = 08** |
| **Marks distribution of the Practical paper** | **End semester = 22**<br>**Internal Marks = 03** |
| **Contact Hours** | **Theory: 3 Hrs. per week**<br>**Practical: 3 Hrs. per week** |

**Applicable to the students**

1. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
2. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

**LEARNING OUTCOMES SPECIFIC TO DISCIPLINE**

By studying this applied course, the students will be able to:

- Explain the details of fats, oils, soap and paper.
- The student shall be able to give the details involved in paper manufacture.
- Explain the working of sugar, cement, cosmetic, perfume and ceramic industries.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

**GENERIC OUTCOMES**

The first year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

**For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.**

**Semester 2: TRAINING/WORKSHOP/INDUSTRIAL VISITS**

1.Visit to a local industry during the year and students to submit a report on the same. The report should contain information about: Raw materials used in industry, Steps involved in manufacturing, Final product and quality control measures involved.

2. Each student to deliver a seminar/power point presentation relevant to theory /practical syllabus.

# Semester-2
# Syllabus for Theory Paper
## ICH-DSC-2-MAJ/MIN-201
## Applied Industrial Chemistry

| | |
|---|---|
| **End semester = 67** | **Credits - 03** |
| **Internal Marks = 08** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

### UNIT-1(15 Hrs.)

**Oils & Fats:** Introduction, Chemistry of oils and fats, including their structure, composition, and classification, Saturated and unsaturated oils/fats, Saturated and unsaturated fatty acids with example, Physical and chemical characteristics of oils/fats, Extraction of oils from vegetable oils, refining of oils and fats, hydrogenation of oils, Determination of quality parameters of oils/fats: acid value, saponification value and iodine value, Industrial applications.

**Soaps and Detergents**: Introduction, Theoretical aspects, Chemistry of soaps and detergents, Chemical composition, structures, and properties of both soaps and detergents, Types of soaps, Various raw materials used in the manufacture of soaps, such as oils, fats, alkalis, and other additives, Manufacture of soap: Cold, semi-boiled and hot process, Cleaning action of soap, Chemical reactions involved in the saponification process, mechanism of action of detergents., Classification of detergents, finishing of detergents, Comparison between Soaps and Detergents.

**Pulp & paper:** Introduction, Raw material preparation, pulping techniques (mechanical and chemical), types of pulp, Manufacture of paper: paper formation and finishing, Chemicals used in paper manufacturing.

### UNIT-2(15 Hrs.)

**Sugar**: Introduction; Sugar industry overview, Chemicals used in sugar processing, Manufacturing of cane sugar, Steps involved in manufacturing, Sugar extraction process: defecation, sulphitation, carbonation, evaporation, concentration, crystallization, clarification, centrifugation, drying, refining; By-Product of sugar industry, Uses of molasses and bagasse.

**Cement:** Introduction to Cement: Definition, classification, and applications, Cement-Types of cement, Constituents of cement, Types of raw materials used in cement production, including limestone, clay, and gypsum, Manufacture of Portland cement, Flow chart of Portland cement manufacturing process, Chemical composition of Portland cement, Mixing and Crushing, Dry Process, Wet Process, Burning Process, Rotary Kiln zones, Grinding and Packaging, Setting and Hardening of Portland cement, Heat of hydration of cement.

**Ceramics**: Introduction, Definition, Properties of ceramics: Chemical, mechanical, thermal, Classification of ceramic materials: Oxides, non-oxides and composites, Traditional classification of ceramics: Pottery, Whiteware, Stoneware, Chinaware, Porcelain, Refractories, Classification of refractories: acid, basic and neutral, Characteristics of raw materials, Properties of refractories, Important steps involved in the manufacture of refractories.

## UNIT-3(15 Hrs.)

**Cosmetics and Perfumes:**Chemistry of cosmetics and perfumes, Introduction to cosmetics and perfumes, Types of cosmetics, Ideal properties of cosmetics, Classification of cosmetics: according to their use, according to their function, according to their physical nature, according to function state.

**General study including preparation and uses**: Hair dye, hair spray, Shampoo.

**Raw material, manufacturing and uses**: Suntan lotions, face powder, lipsticks, talcum powder, nail enamel.

**Preparation and uses**: creams (cold, vanishing and shaving creams), antiperspirants and artificial fragrances.

**Essential oils**: Their importance in cosmetic industries with reference to; Eugenol, Geraniol, Sandalwood oil, Eucalyptus oils and Rose oil, their production and uses.

### Books Recommended

1. Shreev, R.N. & Brink, J.A.: Chemical Process Industries, 5th Edition, McGraw Hill, 1987.
2. Austine, G.T.: Shreeves Chemicals Process Industries, 5th Edition, McGraw Hill, 1984.
3. Dryden, C.E., Rao M.G. & Silting, M.: Outlines of Chemical Technology, 3rd Edition, affiliated East West Press Pvt. Ltd., N. Delhi, 2008.
4. Pandey, G.N.: Chemical Technology, Volume-II, Lion Press, Kanpur.
5. Hench, L.L.: Science of Ceramic chemical processing.
6. Stewarts, G.H.: Science of Ceramics.

### Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

### Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

# Semester-2
# Syllabus for Practical Paper
## ICH-DSC-2-MAJ/MIN-201
## Applied Industrial Chemistry

| | |
|---|---|
| **End semester = 22** | **Credit - 01** |
| **Internal Marks = 03** | **Contact Hours – 3/week** |
| **Examination Time allowed** | **3 Hours** |

**Chemistry Lab: Applied Industrial Laboratory Operations**

1. Determination of Acid value of the given oil/fat material.
2. Determination of Iodine value of the given oil/fat material.
3. To determine Saponification value of the given oil/fat materials.
4. To make a recycled paper on laboratory scale.
5. Carbohydrates: Reducing and Non reducing sugars by:
   (a) Fehling's method (b) Pavy's method
6. Cement: Loss of ignition, silica, insolubles, ash content.
7. To determine the porosity of the given ceramic material.
8. Preparation of nail polish and nail polish remover.
9. Power point presentations.

**Books Recommended**

1. Vermani, O. P.; Narula, A. K. (2004), Industrial Chemistry, Galgotia Publications Pvt. Ltd., New Delhi.
2. Bhatia, S. C. (2004), Chemical Process Industries, Vol. I & II, CBS Publishers, New Delhi.
3. Barel, A.O.; Paye, M.; Maibach, H.I. (2014), Handbook of Cosmetic Science and Technology, CRC Press.
4. Gupta, P.K.; Gupta, S.K. (2011), Pharmaceutics and Cosmetics, Pragati Prakashan.
5. Butler, H. (2000), Poucher's Perfumes, Cosmetic and Soap, Springer.

**General Instructions to the Examiners:**

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

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# Mathematics

## AS PER NEP (2020)

Syllabus of **Mathematics** for UG Classes
Table-1 (SCIENCE TEMPLET)
SINGLE MAJOR, SINGLE / DOUBLE MINOR. **BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

| Sem. | Level | MAJOR (4 Cr) | MINOR (4 Cr) | MDC (3 Cr) Any paper in each of the first three semesters | VAC (2 Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters | SEC (3 Cr) Required three papers, one for each of the first three semesters | AEC (2 Cr) | PBI / HCP (2 Cr) |
|---|---|---|---|---|---|---|---|---|
| I | 100 | • MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry | • MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry | • MAT-MDC-101: Basic Mathematics<br>• MAT-MDC-102: Calculus<br>• MAT-MDC-103: Algebra | • MAT-VAC-101: Vedic Mathematics<br>• MAT-VAC-102: Vedic Algebra<br>• MAT-VAC-103: Theory of Numbers<br>• MAT-VAC-104: Theory of Numbers with html | • MAT-SEC-101: Maths Based Practicals<br>• MAT-SEC-102: Maths Based Programming And Practicals<br>• MAT-SEC-103: Latex And Html<br>• MAT-SEC-104: Quantitative Aptitude<br>• MAT-SEC-105: Discrete Mathematics | Eng -1 | Pbi-1/ HCP-1 |
| II | 100 | • MAT-DSC2 (MAJ/MIN)-201: Calculus-I | • MAT-DSC2 (MAJ/MIN)-201: Calculus-I | | | | Eng -2 | Pbi-2/ HCP-2 |
| III | 200 | • MAT-DSC3 (MAJ/MIN)-301: Theory of Equations<br>• MAT-DSC3 (MAJ)-302: Differential Equations-I | • MAT-DSC3 (MAJ/MIN)-301: Theory of Equations<br>• MAT-DSC3 (MAJ)-302: Differential Equations-I | | | | Eng-3/ MIL-1 | |
| IV | 200 | • MAT-DSC4 (MAJ/MIN)-401: Calculus-II<br>• MAT-DSC4 (MAJ)-402: Discrete Mathematics and Graph Theory<br>• MAT-DSC4 (MAJ)-403: Coordinate Geometry-I<br>• MAT-DSC4 (MAJ)-404: Numerical Analysis | • MAT-DSC4 (MAJ/MIN)-401: Calculus-II | | | | Eng-4/ MIL-2 | |
| V | 300 | • MAT-DSC5 (MAJ/MIN)-501: Linear Algebra<br>• MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers<br>• MAT-DSC5 (MAJ)-503: Differential Equations-II<br>• MAT-DSC5 (MAJ)-504: Solid Geometry<br>• MAT-DSC5 (MAJ)-505: Statics | | | Internship / community work | | | |
| VI | 300 | • MAT-DSC6 (MAJ/MIN)-602: Analysis<br>• MAT-DSC6 (MAJ)-603: Linear Programming<br>• MAT-DSC6 (MAJ)-604: Probability Theory<br>• MAT-DSC6 (MAJ)-605: Dynamics<br>• MAT-DSC6 (MAJ)-606: Modern Algebra | MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry | | | | | |

Table-2 (SCIENCE TEMPLET)
MULTI-DISCIPLINARY / INTER-DISCIPLINARY**. BSc, BSc HONOURS AND BSc HONOURS WITH RESEARCH**

| Sem. | Level | MAJOR (4 Cr) | MINOR (4 Cr) | MDC (3 Cr) Any paper in each of the first three semesters | VAC (2 Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters | SEC (3 Cr) Required three papers, one for each of the first three semesters | AEC (2 Cr) | PBI / HCP (2 Cr) |
|---|---|---|---|---|---|---|---|---|
| I | 100 | • MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry | • MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry | • MAT-MDC-101: Basic Mathematics<br>• MAT-MDC-102: Calculus<br>• MAT-MDC-103: Algebra | • MAT-VAC-101: Vedic Mathematics<br>• MAT-VAC-102: Vedic Algebra<br>• MAT-VAC-103: Theory of Numbers<br>• MAT-VAC-104: Theory of Numbers with html | • MAT-SEC-101: Maths Based Practicals<br>• MAT-SEC-102: Maths Based Programming And Practicals<br>• MAT-SEC-103: Latex And Html<br>• MAT-SEC-104: Quantitative Aptitude<br>• MAT-SEC-105: Discrete Mathematics | Eng -1 | Pbi-1/ HCP-1 |
| II | 100 | • MAT-DSC2 (MAJ/MIN)-201: Calculus-I | • MAT-DSC2 (MAJ/MIN)-201: Calculus-I | | | | Eng -2 | Pbi-2/ HCP-2 |
| III | 200 | • MAT-DSC3 (MAJ/MIN)-301: Theory of Equations | • MAT-DSC3 (MAJ/MIN)-301: Theory of Equations<br>• MAT-DSC3 (MAJ)-302: Differential Equations-I | | | | Eng-3/ MIL-1 | |
| IV | 200 | • MAT-DSC4 (MAJ/MIN)-401: Calculus-II<br>• MAT-DSC4 (MAJ)-402: Differential Equations-I | • MAT-DSC4 (MAJ/MIN)-401: Calculus-II | | | | Eng-4/ MIL-2 | |
| V | 300 | • MAT-DSC5 (MAJ/MIN)-501: Linear Algebra<br>• MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers | • MAT-DSC5 (MAJ/MIN)-501: Linear Algebra | | Internship / community work | | | |
| VI | 300 | • MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry<br>• MAT-DSC6 (MAJ)-602: Analysis | • MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry<br>• MAT-DSC6 (MAJ)-602: Analysis | | | | | |

**Notes.**

**1.** Members of the Board of Studies unanimously decided not to offer mathematics in Table-3 of Arts stream.

**2.** Practicals (wherever included in the syllabi) may be conducted into small groups (say 15 – 20 students) to facilitate the smooth conduct of practical sessions.

**3.** Scientific Calculator is allowed in the Numerical Analysis Paper.

**IMPORTANT NOTE:** All papers in VAC / SEC / MDC would be floated in each of the first three semesters. However, the students are not allowed to repeat any paper.

## Semester - I
## MAT-DSC1 (MAJ/MIN)-101: ALGEBRA & TRIGONOMETRY

**Credits = 4**
**Teaching hours =** 6 hrs / week including tutorials / practicals etc. in a semester of 15 weeks.
**Exam Time:** 3 hours
**Max. Marks:** 100 (External: 80 + Internal Assessment: 20)

***Note:*** 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
**Mid Semester Exam** = 10 marks,
**Assignment / Project work** = 5 marks,
**Classroom participation =** 3 marks**,**
**Attendance =** 2 marks for more than 85% and 1 mark for more than 75% attendance.

**Objective**: The main goal of this course is to deliver basics of Binomial Theorem, Properties of Binomial coefficients . The course introduces the concept of De Moivre's theorem and its applications, eigen values and eigen vectors of a matrix. The objective is to understand nature of solution of a linear system of equations.

**Learning Outcomes**: The students will be able to determine the sum of infinite Binomial Series, the rank of a matrix, eigen values, eigen vectors and solve linear system of equations using matrices.

### UNIT-I

Principle of Mathematical Induction (both strong and weak forms) and its applications (e.g. see [1]). Binomial Theorem for positive index, verification of its correctness by using induction principal, its combinatorial interpretation, Properties of binomial coefficients. Binomial Theorem for any index, summation of infinite binomial series.
Solution of trigonometric equations, sine, cosine and projection formulae for arbitrary triangles (e.g. see [3]).

### UNIT-II

De Moivre's theorem, applications of De Moivre's theorem including primitive $n^{th}$ roots of unity. Expansions of sin $n\theta$, cos $n\theta$ , $sin^n\theta$, $cos^n\theta$ for $n \in$ □. Exponential, logarithmic, circular (direct and inverse) and hyperbolic functions of a complex variable.

### UNIT –III

Determinant of an $n \times n$ matrix and its properties. Definition and properties of hermitian and skew-hermitian matrices. Row and column vectors, linearly dependent and independent vectors, row rank, column rank and their equivalence, rank of a matrix. Rank of product of matrices and rank of sum of matrices.

### UNIT-IV

Theorems on consistency of a system of linear equations (both homogeneous and non-homogeneous). Eigen-values, eigen-vectors and characteristic equation of a matrix, Cayley-Hamilton theorem and its use in finding inverse of a matrix. Diagonalization.

**References:**

1. **I. S. Sominskii** : The Method of Mathematical Induction, Blaisdell Publishing Company New York. London-Pergamon Press Ltd (1961).
2. **David S. Gunderson**: Handbook of Mathematical Induction; Theory and Applications, CRC Press (2010).
3. **C. V. Durell**, **A. Robson** : Advanced Trigonometry, Dover Publications (2003).
4. **K. B. Datta**: Matrix and Linear Algebra, Prentice Hall of India Pvt. Ltd., New Delhi, 2004.
5. **S. R. Knight and H. S. Hall**: Higher Algebra, H. M. Publications, Seventh Edition, 1994.
6. **R. S. Verma and K. S. Shukla**: Text Book on Trigonometry, Pothishala Pvt. Ltd., Allahabad, $10^{th}$ Edition, 1999.
7. **Shanti Narayan and P. K. Mittal**: A Text Book of Matrices, S. Chand & Co., New Delhi, Revised Edition, 2007.

## Semester - II
## MAT-DSC2 (MAJ/MIN)-201: CALCULUS-I

**Credits = 4**
**Teaching hours =** 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
**Exam Time:** 3 hours
**Max. Marks:** 100 (External: 80 + Internal Assessment: 20)

*Note:*
1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
   **Mid Semester Exam** = 10 marks,
   **Assignment / Project work** = 5 marks,
   **Classroom participation =** 3 marks**,**
   **Attendance =** 2 marks for more than 85% and 1 mark for more than 75% attendance.

**Objective**: The main goal of this course is to introduce the basics of limits, continuity and differentiability of functions. Besides this, differentiation of hyperbolic and inverse hyperbolic with their applications will also be studied. The course will cover Mean Value Theorems and their applications.

**Learning Outcomes**: After completing this course the students will be able to compute limits and continuity of various functions. The course will empower the students to compute derivatives of Hyperbolic functions. The students will be able to compute tangents and normals.

### UNIT-I

**Real Numbers**. Order properties of real numbers, bounds, *l.u.b.* and *g.l.b.*, order completeness property of real numbers, Archimedean property of real numbers (e.g. see chapter 1 of [1]) .

**Limits**: Functions (exponential, logarithmic, modulus, trigonometric, polynomials etc.), $\varepsilon$-$\delta$ definition of the limit of a function, basic properties of limits, methods computations, infinite limits (e.g. see chapter 5 of [1]).

### UNIT-II

**Continuity**. $\varepsilon$-$\delta$ definition of a continuous function, various methods to check continuity / discontinuity of a function, types of discontinuities, continuity of composite functions, sign of a function in a neighborhood of a point of continuity, intermediate value theorem, maximum and minimum value theorem.

**Differentiability**. Definition of a differentiable real valued function of a real variable, computing derivatives of elementary functions by using definition. Geometrical meaning of derivative of a function at a point.

### UNIT-III

**Derivatives**. Revision of various rules to compute derivatives (e.g. product rule, quotient rule, chain rule etc.), Introduction to hyperbolic, inverse hyperbolic functions of a real variable, their derivatives. Successive differentiations, Leibnitz's theorem, indeterminate forms.

### UNIT- IV

**Applications of Derivatives**: Tangents and normals, Differentials and Approx./ Errors.

**Mean value theorems**: Rolle's Theorem, Lagrange's mean value theorem, Cauchy's mean value theorem, their geometric interpretation and applications, Taylor's theorem, Maclaurin's theorem with various form of remainders and their applications.

**References**

1. **Brian S. Thomson, Judith B. Bruckner, Andrew M. Bruckner:** Elementary real analysis, Prentice-Hall (2001)
2. **Murray R. Spiegel**: Schaum's outline series Theory and Problems of Advanced Calculus, Schaum Publishing Co. New York, Second Edition, 2002.
3. **P. K. Jain and S. K. Kaushik**: An Introduction to Real Analysis, S. Chand & Co., New Delhi, First Edition, 2000.
4. **G. Prasad**: Differential Calculus, Pothishala Pvt Ltd., Allahabad, Revised Edition,2016.
5. **G. B. Thomas and R. L. Finney**: Calculus and Analytic Geometry, Pearson Publication, Ninth Edition, 2010.
6. **Shanti Narayan and P. K. Mittal**: Differential Calculus, S. Chand & Co., New Delhi, Fifteenth Edition, 1942.
7. **R. G. Bartle, D. R. Sherbert**: Introduction to Real Analysis, Wiley India Pvt. Ltd

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# MICROBIOLOGY

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)**

**B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN MICROBIOLOGY**

| Table1* | | | | | | Table2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** |
| I | L-100 | (MIC-DSC-1 MIN - 101) | Basic Microbiology | 3+1=4 | 24 | I | L-100 | (MIC-DSC-1 MIN - 101) | Basic Microbiology | 3+1=4 | 24 |
| | | DSC- MAJ | To be opted by the student | 3+1=4 | | | | DSC- MAJ | To be opted by the student | 3+1=4 | |
| | | DSC- MAJ | To be opted by the student | 3+1=4 | | | | DSC- MAJ | To be opted by the student | 3+1=4 | |
| | | (MIC - MDC-1-101) | Introduction to Microbiology | 3 | | | | (MIC - MDC-1-101) | Introduction to Microbiology | 3 | |
| | | (VAC-1-101) | To be opted by the student | 2 | | | | (VAC-1-101) | To be opted by the student | 2 | |
| | | (MIC-SEC-1-101) | Tools in Microbiology | 3 | | | | (MIC-SEC-1-101) | Tools in Microbiology | 3 | |
| | | (AEC-1-101) | Eng-1 | 2 | | | | (AEC-1-101) | Eng-1 | 2 | |
| | | (PUN-COMPL- 1 – 101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | | | | (PUN-COMPL- 1 – 101/HCP-COMPL-1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (MIC-DSC-2 MIN - 201) | Microbial Physiology | 3+1=4 | 24 | II | L-100 | (MIC-DSC-2 MIN - 201) | Microbial Physiology | 3+1=4 | 24 |
| | | DSC- MAJ | To be opted by the student | 3+1=4 | | | | DSC- MAJ | To be opted by the student | 3+1=4 | |
| | | DSC- MAJ | To be opted by the student | 3+1=4 | | | | DSC- MAJ | To be opted by the student | 3+1=4 | |
| | | (MIC-MDC-2-201) | Tools and Techniques in Microbiology | 3 | | | | (MIC-MDC-2-201) | Tools and Techniques in Microbiology | 3 | |
| | | (VAC-2-201) | To be opted by the student | 2 | | | | (VAC-2-201) | To be opted by the student | 2 | |
| | | (MIC-SEC-2-201) | Biosafety in Microbiology | 3 | | | | (MIC-SEC-2-201) | Biosafety in Microbiology | 3 | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | (AEC-2-201) | Eng-2 | 2 | |
| | | (PUN-COMPL- 2 - 201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | | | | (PUN-COMPL- 2 - 201/HCP-COMPL-2-201) | Pbi-2/HCP-2 | 2 | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II | | | | | | | | | | | |
| The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/communitywork. | | | | | | | | | | | |

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## SEMESTER-I

### MIC-DSC-1 MIN – 101:Basic Microbiology

**Total marks: 75**
**Theory: 67**
**Internal assessment: 08**
**Time: 3 hours**
**Credits: 3**

**Note:**The question paper will consist of four Units (I-IV).There will be nine questions, out of which five questions haveto be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be two questions from each Unit and one is to be attempted from each Unit.Each question should be sub-divided into 2-4 sub parts.

**Objective:** Aim of the course is to introduce the students with microbial world and to make the students to learn the morphology, microscopy, staining.

**Learning Outcome:** After the successful completion of the course, the students will be able to understandthe techniques of microscopy, staining and morphology of the bacteria. It will also enrich the students' knowledge on the development of microbiology as the discipline.

### UNIT-I

**NO. OF HOURS: 15**

**HISTORY OF MICROBIOLOGY**
Development of microbiology as a discipline, Spontaneous generation vs. biogenesis. Contributions of Antoni van Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming. Role of microorganisms in fermentation, Germ theory of disease.

**MICROSCOPY**
Microscopic examination of micro-organism, bright field microscopy, dark field microscopy, phase contrast microscopy and electron microscopy.

**STAINING**
Nature of stains, principles, mechanism, methods and types of staining: simple, gram staining, capsule staining and acid fast staining.

### UNIT-II

**NO. OF HOURS: 15**

**MORPHOLOGY AND FINE STRUCTURE OF BACTERIA**
Cell size shape and arrangement, Organization of cell wall, cell membrane, flagella and capsules, fimbriae and pili, mesosomes, nucleoid chromosome and plasmids in bacteria.

**BACTERIOLOGICAL TECHNIQUES**
Pure culture isolation: Culture media, Streaking, serial dilution and planting methods; cultivation, maintenance and preservation/stocking of pure cultures; cultivation of anaerobic bacteria.

## UNIT-III

**NO. OF HOURS: 15**

### MICROBIAL GROWTH

Phases of growth, generation time, growth rate, bacterial growth curve, mathematical nature and expression, diauxic growth, chemostat and turbidostatic. Factors affecting growth in microorganisms. Methods for measurement of growth in microorganisms, formation of endospores in bacteria.

## UNIT-IV

**NO. OF HOURS: 15**

### CONTROL OF MICROORGANISMS

Physical and chemical methods of sterilization, disinfection, antimicrobial agents, sulfa drugs, antibiotics: broad-spectrum antibiotics, beta-lactam antibiotics.

### NORMAL MICROBIAL FLORA

Germfree and gnotobiotic life, distribution and occurrence of the normal flora : skin, eye, respiratory tract, mouth, intestinal tract, genitourinary tract.

## MIC-DSC-1 MIN – 101: Basic Microbiology (PRACTICALS)

**Total marks: 25**
**Practical: 22**
**Internal assessment: 03**
**Time: 3 hours**
**Credits: 1**

**List of the Practical:**

1. Preparation of different media: Synthetic Media, Complex media (Nutrient Agar, McConkey agar).
2. Simple staining.
3. Negative staining.
4. Capsule staining.
5. Isolation of pure cultures of bacteria by streaking method.
6. Estimation of CFU count by spread plate method/pour plate method.
7. Demonstration of Motility by hanging drop method.

**SUGGESTED READINGS:**

1. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall.
2. Madigan MT, and Martinko JM. (2014). Brock Biology of Micro-organisms. 14thedition.
3. ParkerJ.Prentice Hall International, Inc.
4. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
5. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
6. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.
7. Tortora GJ, Funke BR, and Case CL. (2008). Microbiology:An Introduction. 9th edition Pearson Education.
8. Willey JM,Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Cappucino J and Sherman N.(2010).Microbiology:A LaboratoryManual9$^{th}$edition.Pearson Publication

## SEMESTER-II

### MIC-DSC-2 MIN – 201: Microbial Physiology

**Total marks: 75**
**Theory: 67**
**Internal assessment: 08**
**Time: 3 hours**
**Credits: 3**

**Note**: Question paper will have four Units(I-IV). Examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus. The students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

**Objectives:** The course gives an overview of structure, biological importance of carbohydrates, lipids, biological enzymes, proteins and their properties. Also to understand DNA and RNA ,types of RNA and their biological functions.

**Learning Outcome:** By the conclusion of this course, the students will be able to understand the concepts of carbohydrates, lipids and proteins and also the concepts involved in DNA replication , structure of enzymes and various catabolic and anabolic pathways.

### UNIT-I

**NO. OF HOURS: 15**

**Carbohydrates:**
Monosaccharides: Classification based on aldehyde and ketone groups; structure of Ribose, Deoxyribose, Glucose, Galactose and Fructose.
Disaccharides: Glycosidic bond, structure of lactose and sucrose.
Polysaccharides : Structure and biological role of starch, glycogen and cellulose.

**Lipids :**
1.Simple lipids – Fats and oils, waxes. 2. Compound lipids – Phospholipid, Glycolipids 3. Derived lipids – Cholesterol

**Enzymes:**
1. Definition, 2. Structure- Concept of apoenzyme, coenzyme, cofactor 9 and active site. 3. Types- Extracellular, Intracellular, Constitutive and Inducible. 4. Features of enzyme - substrate reaction.

### UNIT-II

**NO. OF HOURS: 15**

**Nucleic Acids:**
DNA – structure and composition (Watson and Crick Model) 2. RNA – Types (m-RNA, t-RNA, r-RNA), structure and functions.

**Proteins:**
Structural levels of proteins: primary, secondary, tertiary and quaternary domain structure,protein classification and function.

## UNIT-III

**NO. OF HOURS: 15**

**Enzymes:**
Chemical and physical properties of enzymes. Classification and nomenclature of Enzymes. Mechanisms of enzymes action. Inhibition, control and regulation of enzyme activity. Factors affecting enzyme activity. Enzyme kinetics: Michaelis-Menten equation.

## UNIT-IV

**NO. OF HOURS: 15**

**ATP generation:**
Substrate level phosphorylation, Oxidative phosphorylation, Bacterial Photophosphorylation – Cyclic and Non- cyclic, Catabolism of glucose – EMP, Tricarboxylic Acid Cycle, Pentose Phosphate Pathway, Entner Doudoroff Pathway.

## MIC-DSC-2 MIN – 201: Microbial Physiology(Practicals)

**Total marks: 25**
**Practical: 22**
**Internal assessment: 03**
**Time: 3 hours**
**Credits: 1**

**List of Experiments:**

1. Qualitative Analysis of Carbohydrates.
2. Colorimetric estimation of proteins by Biuret / Lowry method.
3. Separation of mixtures by paper / thin layer chromatography.
4. Effectoftemperature/pH / Salt concentration on bacterial growth.
5. Demonstration of electrophoretic technique.
6. Study and plot the growth curve of E. coli by turbidometric and Standard Plate Count methods.

**SUGGESTED READINGS:**

1. Berg JM, Tymoczko JL and Stryer L (2011) Biochemistry, W.H.Freeman and Company Caldwell, D.R. (1995). Microbial Physiology and Metabolism, W.C. Brown Publications,Iowa, USA.
2. Sashidhara Rao, B. and Deshpande, V. (2007). Experimental Biochemistry: A student Companion. I.K. International Pvt. Ltd.
3. Tymoczko JL, Berg JM and Stryer L (2012) Biochemistry: A short course, 2nd ed.
4. W.H.Freeman 6. Voet,D. and Voet J.G (2004) Biochemistry 3rd edition, John Wiley and Sons
5. Fisher, M. (2001). Lehninger principles of biochemistry, ; by David L. Nelson and Michael M. Cox. The Chemical Educator, 6, 69-70

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# Physics

**NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)**

**B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH**

| Table 1* | | | | | | Table 2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | Total Credits | Sem | Level | Course Code | Course Name | Credits (Th+Pr+Tut) | Total Credits |
| I | L-100 | (PHY-DSC-1 Maj/Min-101) | Mechanics | 3+1=4 (x3) | 24 | I | L-100 | (PHY-DSC-1 Maj/Min-101) | Mechanics | 3+1=4 (x3) | 24 |
| | | (PHY-MDC-1-101) | Green Energy | 3 | | | | (PHY-MDC-1-101) | Green Energy | 3 | |
| | | (PHY-VAC-1-101) | Science in Everyday Wonders | 2 | | | | (PHY-VAC-1-101) | Science in Everyday Wonders | 2 | |
| | | (PHY-SEC-1-101) | Physics instrumentation Skills-I | 3 | | | | (PHY-SEC-1-101) | Physics instrumentation Skills-I | 3 | |
| | | (AEC-1-101) | Eng-1 | 2 | | | | (AEC-1-101) | Eng-1 | 2 | |
| | | (PUN-COMPL1-101/HCP-COMPL1-101) | Pbi-1/HCP-1 | 2 | | | | (PUN-COMPL1-101/HCP-COMPL1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (PHY-DSC-2 Maj/Min-201) | Electricity and Magnetism | 3+1=4 (x3) | 24 | II | L-100 | (PHY-DSC-2 Maj/Min-201) | Electricity and Magnetism | 3+1=4 (x3) | 24 |
| | | (PHY-MDC-2-201) | Dynamics of Motion and Human Health | 3 | | | | (PHY-MDC-2-201) | Dynamics of Motion and Human Health | 3 | |
| | | (PHY -VAC-2-201) | Fundamentals of Solar Energy and Applications | 2 | | | | (PHY -VAC-2-201) | Fundamentals of Solar Energy and Applications | 2 | |
| | | (PHY -SEC-2-201) | Physics instrumentation Skills-II | 3 | | | | (PHY -SEC-2-201) | Physics instrumentation Skills-II | 3 | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | (AEC-2-201) | Eng-2 | 2 | |
| | | (PUN-COMPL2-201/HCP-COMPL2-201) | Pbi-2/HCP-2 | 2 | | | | (PUN-COMPL2-201/HCP-COMPL2-201) | Pbi-2/HCP-2 | 2 | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester | | | | | | | | | | | |
| The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work. | | | | | | | | | | | |

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attainatleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## Semester 1

## PHY-DSC-1 Maj/Min-101
## (Mechanics)

**Maximum marks: 100**
**(Theory 75 marks & Practical 25 marks)**
**Theory 75 marks [60 marks+15 marks (internal assessment 20%)]**
**Practical 25 marks [20 marks+5 marks (internal assessment 20%)]**
**Teaching hours: Theory- 3Hrs per week**
**Total Credits: 3(Theory)+1(Practical)**

### Learning Outcomes

- To develop a comprehensive understanding of Cartesian and spherical polar coordinate systems and explore the concept of solid angles and their significance in physical problems.
- To establish the relationship between the conservation laws and the symmetries of space and time.
- To explore elastic collisions in laboratory and center-of-mass reference frames and to derive relationships between velocities, angles, and kinetic energies in these frames.
- To understand Galilean transformations, invariance, and the transformation equations for inertial frames inclined to each other.
- To investigate the postulates of the special theory of relativity and explore the kinematical consequences of Lorentz transformations, including length contraction, time dilation, and the twin paradox.
- To derive and understand the relativistic transformation of velocities, the concept of simultaneity in relativity, and the relativistic Doppler effect.

**Note:**

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

### UNIT-I

Cartesian, plane polar and spherical polar co-ordinate systems, area, volume, displacement, velocity, and acceleration in these systems, Relationship of conservation laws of linear momentum, angular momentum and energy, symmetries of space and time. Various forces in nature, relative strengths and spatial dependence, Motion under force obeying inverse square law, equivalent one-body problem. Motion under central forces, equation of motion under central force, equation of orbit and turning points, Kepler's Laws. Elastic collision in lab and C.M. systems, relationships of velocities, angles, and kinetic energies in these two systems.

## UNIT-II

Galilean transformations and Invariance, Non-Inertial frames. Fictitious forces in a rotating frame of reference, Centrifugal and Coriolis forces due to rotation of earth.Concept of stationery universal frame of reference and ether, Michelson-Morley experiment, and its results.

Postulates of special theory of relativity, Lorentz transformations, Kinematical consequences of Lorentz transformations – length contraction and time dilation, Twin paradox, Transformation of velocities,Relativistic Doppler effect.Variation of mass with velocity, mass-energy equivalence.

**Books Suggested:**

**Essential Readings:**

1. Mechanics, Berkeley, Vol. I, C. Kittle.
2. Mechanics, H.S. Hans & S.P. Puri.

**Further Readings:**

1. An Introduction to Machines, Daniel Kleppner& Robert J. Kolenkow (TMH).
2. Introduction of Classical Mechanics, R.G. Takwale&P.S.Puranik(TMH,2000).
3. Mechanics & Relativity (3rd Edition), Vidwan Singh Soni (PHI Learning, New Delhi, 2013)
4. Basic Concepts of Relativity, R.H. Good (East-West Press, New Delhi, 1974).

# PHYSICS PRACTICALS
## (PHY-DSC-1 Maj/Min-101)

**Max. Marks: 25 (20+5 internal assessment)** **Total Credit: 1**
**Teaching hours: 2Hrs per week**

**General Guidelines for Physics Practical Examinations:**

**1.** The distribution of marks is as follows:

| | |
|---|---|
| (i) One full experiment requiring the student to take some data, analyze it and draw conclusions.(Candidates are expected to state their results with limits of error). | 09 |
| (ii) One exercise based on experiment | 04 |
| (iii) Viva-Voce | 04 |
| (iv) Record (Practical file) | 03 |
| (v) Internal Assessment | 05 |

***Note for Examiners:***

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

**Analysis of Experimental Data (Compulsory for all students in first semester):**

**Objectives:**

(i) Knowledge of propagation of errors.
(ii) Knowledge of significant figures, Determination of standard deviation and probable error and their use in interpretation of the experimental result.
(iii) Familiarity with the method of least square fitting of experimental data to a curve.

**LIST OF EXPERIMENTS:**

1. Todetermineenergy transfer, coefficient of restitution and verify laws of conservation of linear momentum and kinetic energy in elastic collisions using one dimensional collision of hanging spheres.
2. To find out the moment of inertia of a flywheel.
3. To establish relationship between torque and angular acceleration using fly wheel.
4. Measure time-period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.
5. To find Young's Modulus of the material of the given beam by bending using a travelling microscope.
6. To Determine Poisson's ratio for rubber.
7. Find the value of g by Katers pendulum.

8. Measure time period as a function of distance of centre of suspension (oscillation) from centre of mass, plot relevant graphs, determine radius of gyration and acceleration due to gravity.
9. Determination of coefficient of viscosity of a given liquid by Stoke's method.
10. Basics of measurements with Vernier calliper, screw gauge, spherometer.

**Texts and Reference Books:**

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3. "Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

## Semester 2
## PHY-DSC-2 Maj/Min-201

## (ELECTRICITY AND MAGNETISM)

**Maximum marks: 100**
**(Theory 75 marks & Practical 25 marks)**
**Theory 75 marks [60 marks+15 marks (internal assessment 20%)]**
**Practical 25 marks [20 marks+05 marks (internal assessment 20%)]**
**Teaching hours: Theory- 3 Hrs per week**
**Total Credits: 3(Theory)+1(Practical)**

### Learning Outcomes

- To gain a solid foundation in vector calculus, including the concepts of gradient, divergence, and curl in Cartesian coordinates.
- To study the fundamental principles of electrostatics, including Coulomb's law, electric fields from various charge distributions, and the concepts of electric potential and work.
- To explore the behavior of materials in electric fields, focusing on polarization, dielectric properties, and the relationships between electric susceptibility, permittivity, and the displacement vector.
- To study the behavior of electric currents, including current density and the microscopic form of Ohm's law.
- To understand the response of materials to magnetic fields, including the concepts of magnetization, magnetic susceptibility, permeability, and the relationships between free and bound currents.

**Note:**

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination Centre.
5. The Logarithmic table will be provided in the examination centre, if required.

### UNIT-I

Basic ideas of Vector Calculus, Del Operator, Gradient, Divergence, curl in Cartesian coordinates, physical significance and applications, Conservative field, Laplacian in Rectangular coordinates. Stoke's theorem, Gauss's divergence theorem, Coulomb's Law for point charges and continuous distribution of charges, electric field due to dipole, line charge, charged ring, circular disc, spherical shell, solid sphere, Gauss's Law and its differential form, Applications of Gauss's Law.

Work and potential difference, Potential difference as line integral of field, Electric potential due to uniformly charged wire, circular ring, spherical shell, dipole, linear quadrupole, Electric field as gradient of scalar potential, curl E = 0. Poisson and Laplace's equation.

Polarisation of matter, atomic and molecular dipoles, induced dipole moment and atomic polarizability. Electric susceptibility and polarization vector. Relation $K=1+\chi$, Gauss's law for dielectrics. Displacement vector, Div. D = 0.

## UNIT-II

Current density and equation of continuity. Microscopic form of Ohm's Law ($\mathbf{J} = \sigma\mathbf{E}$). Invariance of charge, **E**in different frames of reference.

Behaviour of various substances in magnetic field. Definition of M and H, Permeability and susceptibility and their inter relationship.

Lorentz's force. Definition of **B**. Biot Savart's Law and its application to long straight wire, circular current loop and solenoid. Ampere's Circuital law and its applications. Divergence and curl of **B**. Faraday's Law of EM induction, Self and Mutual inductance and reciprocity theorem.

**Books Suggested:**

**Essential Readings:**

1. Electricity & Magnetism, A.S. Mahajan& A.A. Rangwala (Tata McGraw Hill).
2. Fundamentals of Electricity and Magnetism by Arthur F. Kipp.
3. Electricity and Magnetism, Berkeley Physics Course, Vol. II by E.M. Purcell.
4. Introduction to Classical Electrodynamics by David Griffith, Prentice Hall.

**Further Readings:**

1. Electricity & Magnetism, 4th Edition, W.J. Duffin.
2. EM Waves and Radiating Systems, Edward C. Jordan and K. G. Balmain, Prentice Hall.

# PHYSICS PRACTICALS
## (PHY-DSC-2 Maj/Min-201)

**Max. Marks: 25 (20+5 internal assessment)** **Total Credit: 1**
**Teaching hours: 2Hrs per week**

**General Guidelines for Physics Practical Examinations:**

1. The distribution of marks is as follows:

| | |
|---|---|
| (i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error). | 09 |
| (ii) One exercise based on experiment | 04 |
| (iii) Viva-Voce | 04 |
| (iv) Record (Practical file) | 03 |
| (v) Internal Assessment | 05 |

***Note for Examiners:***

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

**LIST OF EXPERIMENTS:**

1. To study the working of household energy meter.
2. To study the induced e.m.f. as function of the velocity of the magnet.
3. To determine low resistance with Carey Fosters Bridge.
4. To study the magnetic field produced by a current carrying solenoid using a search coil and calculate permeability of air.
5. Study of phase relationships using impedance triangle for LCR circuit.
6. To determine the capacitance of a capacitor using flashing and quenching of a neon lamp.
7. To determine the value of air capacitance by de-Sauty's method and hence to find out the permittivity of air.
8. To determine the coefficient of self-induction of a coil (L) using Anderson Bridge.
9. To determine the force on conductor carrying current in a magnetic field.

**Texts and Reference Books :**

1. "B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.
3."Programming with C, Schaum series" by Byron Gottfried&JitenderChhabra.

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# STATISTICS

## NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

## B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

| Table 1* | | | | | | Table 2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** | **Sem** | **Level** | **Course Code** | **Course Name** | **Credits (Th+Pr+Tut)** | **Total Credits** |
| I | L-100 | (STAT-DSC-1 Maj/Min-101) | Descriptive Statistics-I | 3+1=4 (x3) | 24 | I | L-100 | (STAT-DSC-1 Maj/Min-101) | Descriptive Statistics-I | 3+1=4 (x3) | 24 |
| | | (STAT-MDC-1-101) | Statistics-I | 3 | | | | (STAT-MDC-1-101) | Statistics-I | 3 | |
| | | (VAC-1-101)* | -- | 2 | | | | (VAC-1-101)* | -- | 2 | |
| | | (STAT-SEC-1-101) | Statistical Data Analysis using MS-Excel | 2+1=3 | | | | (STAT-SEC-1-101) | Statistical Data Analysis using MS-Excel | 2+1=3 | |
| | | (AEC-1-101) | Eng-1 | 2 | | | | (AEC-1-101) | Eng-1 | 2 | |
| | | (Pbi/HCP-1-101) | Pbi-1/HCP-1 | 2 | | | | (Pbi/HCP-1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (STAT-DSC-2 Maj/Min-201) | Probabiltiy Theory | 3+1=4 (x3) | 24 | II | L-100 | (STAT-DSC-2 Maj/Min-201) | Probabiltiy Theory | 3+1=4 (x3) | 24 |
| | | (STAT-MDC-2-201) | Statistics-II | 3 | | | | (STAT-MDC-2-201) | Statistics-II | 3 | |
| | | (VAC-2-201)* | --- | 2 | | | | (VAC-2-201)* | --- | 2 | |
| | | (STAT-SEC-2-201) | Statistics for Decision Analytics with Microsoft Excel | 2+1=3 | | | | (STAT-SEC-2-201) | Statistics for Decision Analytics with Microsoft Excel | 2+1=3 | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | (AEC-2-201) | Eng-2 | 2 | |
| | | (Pbi/HCP-2-201) | Pbi-2/HCP-2 | 2 | | | | (Pbi/HCP-2-201) | Pbi-2/HCP-2 | 2 | |
| The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester | | | | | | | | | | | |
| The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work. | | | | | | | | | | | |
| ***Value Added Courses are not being offered by the Deaprtment of Statistics, Panjab University, Chandigarh** | | | | | | | | | | | |

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

## Semester-I

Course Code: (STAT-DSC-1 Maj/Min-101)
Course Name: Descriptive Statistics-I

**Credit: 4**
**L-3, P-1**

**Notes:**

1. A candidate shall offer this subject in B.Sc. (Hons) only if he/she studied Mathematics as a subject at 10+2 level.
2. Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
3. Practical examination will be held in the respective semester.

| Paper Title with code | Lecture(L)/ Credit | Practical(P)/ Credit | Marks Distribution | |
|---|---|---|---|---|
| Descriptive Statistics-I<br><br>(STAT-DSC-1 Maj/Min-101) | Lecture :3<br>Credit: 3 | Practical<br>Lecture:2<br>Credit:1 | Theory | Practical |
| | | | Max. Marks: 75<br><br>Theory: 60<br>Internal Assessment : 15<br>Time: 3 Hours | Max. Marks: 25<br><br>Viva voce: 5<br>Record of the practical: 5<br>Written practical Exam: 15<br>Time: 3 Hours |

**DESCRIPTIVE STATISTICS-I**

**Objective:** The objective of the course is to Gain proficiency in presenting data through frequency distribution, diagrams, and graphical representations. To apply descriptive statistical techniques/methods to real-world datasets and draw meaningful insights.The focus will be both on theoretical as well as practical approach based on descriptive statistics.

**Learning Outcomes:** On completion of the course, the student will be able to:

1. Understand the origin, development, scope and importance of statistics.
2. Understand the nature of statistical data.
3. Identify the need of Classification and Tabulation.
4. Understand moments and their relationships, including their effects on change of origin and scale.
5. Construct frequency tables, interprets the data, and identifies the importance of diagrammatic and graphical presentation of data.
6. Explain, evaluate and interpret various measures of central tendency, dispersion, skewness and kurtosis.

**Note**: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

**UNIT-I**

**Introduction of Statistics**: Origin, development, definition, scope, uses and limitations.

**Types of Data**: Qualitative and quantitative data, nominal and ordinal data, time series data, discrete and continuous data, frequency and non-frequency data.

**Collection and Scrutiny of Data**: Collection of primary and secondary data- its major sources including some government publications, scrutiny of data for internal consistency and detection of errors of recording, classification and tabulation of data.

**UNIT-II**

**Presentation of Data**: Frequency distribution and cumulative frequency distribution, diagrammatic and graphical presentation of data, construction of bar, pie diagrams, histograms, frequency polygon, frequency curve,ogives, stem-and-leaf-chart, Box and whisker plot.

**Measures of Central Tendency or Averages**: Concepts of central tendency, characteristics for an ideal measure of central tendency, arithmetic mean, median, mode, geometric mean, harmonic mean; partition values-quartiles, deciles, percentiles and their graphical location along with their properties, applications, merits and demerits.

**UNIT-III**

**Measures of Dispersion**: Concept of dispersion, characteristics for an ideal measure of dispersion. Absolute and relative measures based on: range, inter quartile range, quartile deviation, coefficient of quartile deviation, Mean deviation, coefficient of mean deviation, variance, standard deviation ($\sigma$), coefficient of variation and properties of these measures, root mean square deviation and their relationship, variance of the combined series.

Moments: Moments about mean and about any point and derivation of their relationships, effect of change of origin and scale on moments, Sheppard's correction for moments (without derivation), Charlier's checks; Pearson's $\beta$ and $\gamma$ coefficients.

**UNIT-IV**

Skewness and Kurtosis: Concept of skewness, its types, Absolute and relative (coefficient) measures of skewness with their interpretations and concept of Kurtosis, its types and coefficients of kurtosis with their interpretations.

**Theory of Attributes**: Symbolic notations, dichotomy of data, class frequencies, order of class frequencies, consistency of data, independence and association of attributes, Yule's coefficient of association and coefficient of colligation and their relationship.

**Books recommended**

| S. No. | Title of Book | Name of author | Publisher |
|---|---|---|---|
| 1. | Fundamental of Statistics Vol. I | Goon A.M., Gupta M.K., Dasgupta B. | World Press, Calcutta |
| 2. | Statistics | Johnson R. | Wiley Publishers |
| 3. | Basic Statistics International | Aggarwal B.L. | New Age |
| 4. | Fundamentals of Mathematical Statistics | Gupta S.C.& Kapoor V.K. | Sultan Chand & Sons |

(STAT-DSC-1Maj/Min-101):**DESCRIPTIVE STATISTICS-I (PRACTICAL)**

**Notes:**

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-1Maj/Min-101) will be based on the following topics:

1. To construct frequency distributions using exclusive and inclusive methods
2. Representation of data using Bar and pie diagrams
3. Representation of data using Histogram, Frequency Polygon, Frequency Curve and Ogives.
4. Graphs: stem-and-leaf-display and Box and Whisker Plot
5. To compute various measures of central tendency.
6. To compute various measures of dispersion
7. To obtain first four moments for the given grouped frequency distribution.
8. To obtain various coefficients of skewness and kurtosis.
9. To discuss the association of attributes for a 2x2 contingency table using Yule's coefficient of association and colligation.

## Semester-II

Course Code: (STAT-DSC-2Maj/Min-201)
Course Name: Probability Theory

**Credit: 4**
**L-3, P-1**

**Notes:**

1. Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
2. Practical examination will be held in the respective semester.

| Paper Title with code | Lecture(L)/ Credit | Practical(P)/ Credit | Marks Distribution | |
|---|---|---|---|---|
| Probability Theory<br><br>(STAT-DSC-2 Maj/Min-201) | Lecture :3<br>Credit: 3 | Practical Lecture:2<br>Credit:1 | Theory | Practical |
| | | | Max. Marks: 75<br><br>Theory: 60<br>Internal Assessment : 15<br>Time: 3 Hours | Max. Marks: 25<br><br>Viva voce: 5<br>Record of the practical: 5<br>Written practical Exam: 15<br>Time: 3 Hours |

**PROBABILITY THEORY**

**Objective**: The main objective of the course is to learn different approaches to defining probability methods. Apply probability concepts and techniques to analyze and solve problems in various fields. Also, the aim is to familiarize with random variables, probability functions, and their distributions, including discrete and continuous cases.
Gain proficiency in generating functionsand understand their properties. The focus will be on theoretical as well as practical approach
.

**Learning Outcomes:** On completion of this course students will be able to:

- Understand the concepts of probability, random variables and independence of random variables.
- Explain and apply various definitions and approaches to probability including classical, relative frequency, statistical, and axiomatic approaches.
- Understand and apply moments generating function, cumulant generating function, and probability generating function.
- Utilize generating functions to compute moments and cumulants.
- Analyze functions of random variables and compute joint, marginal, and conditional probability distribution functions.
- Apply different types of probability approaches to solve problems.

**Note**: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

**UNIT-I**

**Concepts in Probability:** Random experiment, trial, sample point, sample space, operation of events, exhaustive, equally likely, mutually exclusive and independent events; Definition of probability-classical, relative frequency, statistical and axiomatic approach.

**UNIT-II**

Conditional probability. Addition and multiplication laws of probability and their extension to *n* events. Boole's inequality; Baye's theorem and its applications.

**UNIT-III**

**Random Variable and Probability Functions:** Definition of random variable, discrete and continuous random variable, probability function, probability mass function and probability density functions, distribution function and its properties, functions of random variables, joint, marginal and conditional probability distribution function.

**UNIT-IV**

**Mathematical Expectation**: Definition and its properties-moments, addition and multiplication theorem of expectation. Conditional expectation and conditional variance.

**Generating Functions:** Moments generating function, cumulant generating function, probability generating function along with their properties.Characteristic function.

**Books recommended**

| S. No. | Title of Book | Name of author | Publisher |
|---|---|---|---|
| 1. | Fundamentals of Mathematical Statistics | Gupta S.C.& Kapoor V.K. | Sultan Chand & Sons |
| 2. | Elementary Probability | David S. | Oxford Press |
| 3. | Introduction to the Theory of Statistics | Mood A.M, Graybill F.A and Boes D.C. | McGraw Hill. |
| 4. | Introduction to Mathematical Statistics | Kapoor&Saxena. | S.Chand |

## (STAT-DSC-2 Maj/Min-201): PROBABILITY THEORY **(PRACTICAL)**

**Notes:**

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-2 Maj/Min-201)will be based on the following topics:

1. Problems based on classical, statistical and axiomatic definition of probability.
2. Problems based on addition and multiplication law of probability.
3. Problems based on conditional and Baye's theorem of probability.
4. Problems based on univariate random variables and distribution function.
5. Problems based on marginal and conditional probability distribution function.
6. Problems based on mathematical expectation.
7. Problem based on generating functions.

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# Zoology

## STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table1*) AND STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY (Table 2*)

## B.Sc.,B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN ZOOLOGY

| Table1* | | | | | | Table2* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sem** | **Level** | **CourseCode** | **CourseName** | **Credits Th(L) + Pr(P)** | **Total Credits** | **Sem** | **Level** | **CourseCode** | **CourseName** | **Credits Th(L) + Pr(P)** | **Total Credits** |
| I | L-100 | (ZOO-DSC-1 Maj/Min-101) | Diversity of Non-chordates | 3+1=4 (x3) | 24 | I | L-100 | (ZOO-DSC-1 Maj/Min-101) | Diversity of Non-chordates | 3+1=4 (x3) | 24 |
| | | (ZOO-MDC-1-101) | Health and Hygiene | 3+0=3 | | | | (ZOO-MDC-1-101) | Health and Hygiene | 3+0=3 | |
| | | (ZOO-VAC-1-101) | Basic Laboratory Techniques | 1+1=2 | | | | (ZOO-VAC-1-101) | Basic Laboratory Techniques | 1+1=2 | |
| | | (ZOO-SEC-1-101) | Vermiculture and Vermicomposting | 2+1=3 | | | | (ZOO-SEC-1-101) | Vermiculture and Vermicomposting | 2+1=3 | |
| | | (AEC-1-101) | Eng-1 | 2 | | | | (AEC-1-101) | Eng-1 | 2 | |
| | | (PUN-COMPL-1-101/HCP- COMPL-1-101) | Pbi-1/HCP-1 | 2 | | | | (PUN-COMPL-1-101/HCP- COMPL-1-101) | Pbi-1/HCP-1 | 2 | |
| II | L-100 | (ZOO-DSC-2 Maj/Min-201) | Diversity of Chordates | 3+1=4 (x3) | 24 | II | L-100 | (ZOO-DSC-2 Maj/Min-201) | Diversity of Chordates | 3+1=4 (x3) | 24 |
| | | (ZOO-MDC-2- 201) | Understanding Human Body | 3+0=3 | | | | (ZOO-MDC-2- 201) | Understanding Human Body | 3+0=3 | |
| | | (ZOO-VAC-2-201) | Vaccines and Vaccination | 1+1=2 | | | | (ZOO-VAC-2-201) | Vaccines and Vaccination | 1+1=2 | |
| | | (ZOO-SEC-2-201) | Apiculture | 2+1=3 | | | | (ZOO-SEC-2-201) | Apiculture | 2+1=3 | |
| | | (AEC-2-201) | Eng-2 | 2 | | | | (AEC-2-201) | Eng-2 | 2 | |
| | | (PUN-COMPL-2-201/HCP- COMPL-2-201) | Pbi-2/HCP-2 | 2 | | | | (PUN-COMPL-2-201/HCP- COMPL-2-201) | Pbi-2/HCP-2 | 2 | |
| **The candidate has to select Major and Minor out of 3 subjects studied in the Semesters I and II at the end of the Semester II** | | | | | | | | | | | |
| **The student exiting the programme after one year, securing 48 credits, will be awarded UGC ertificate provided he/she secures 4 credits from summer internship/community work.** | | | | | | | | | | | |

***Note:The candidate will opt for either Template 1 or 2 from Semester III onwards.**

**Notes:**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

## Semester- I (L-100)

**Course Code: ZOO-DSC-1Maj/Min-101**
**Course Title: Diversity of Non-Chordates**

**Total Hours= 75 (Theory= 45 + Practical= 30)** **Credits= 4 (L=3 + P=1)**
**Total Marks= 100 (Theory=75 + Practical=25)**
**Theory Marks= 75** {67 (Theory examination) +08 (Internal assessment)}
**Practical Marks= 25** {22 (Practical examination) +03 (Internal assessment)}
**Duration for final examination: 3 hours**

### Theory

*Learning Outcomes:*

- *The students will be able to develop an appreciation for the biodiversity of invertebrate species.*
- *The students will acquire knowledge about co-existence of different forms of living organisms ranging from unicellular to multicellular animals.*
- *The students will be able to identify different non-chordates in their surroundings.*
- *The students will be able to apply their knowledge towards identification of pathogenic non-chordates and their management.*

**UNIT- 1**

**Protozoa**: General characteristics and classification up to classes; Eco-Morphological notes on *Euglena, Amoeba*, *Paramecium &Plasmodium.*
Locomotion and reproduction in Protozoans.
**Porifera**: General characteristics and classification up to classes; Eco-Morphological notes on *Sycon, Hyalonema, Euplectella&Spongilla.*
Canal system and Spicules in sponges.

**UNIT- 2**

**Cnidaria***:* General characteristics and classification up to classes; Eco-Morphological notes on *Obelia, Hydra,Physalia, Millepora, Aurelia, Tubipora, Alcyonium, Gorgonia, Metridium&Fungia.*
Polymorphism in *Cnidaria*.
Coral reefs: Types, Formation and Economic Importance.
**Platyhelminthes***:* General characteristics and classification up to classes;Eco-Morphological notes on *Fasciola hepatica &Taeniasolium*
**Nemathelminthes***:* General characteristics and classification up to classes; Eco-Morphological notes on*Ascaris&Ancylostoma.*
Adaptations of Helminth parasites.

**UNIT- 3**

**Annelida:** General characteristics and classification up to classes; Eco-Morphological notes on *Nereis, Chaetopterus, Pheretima&Hirudinaria.*
Metamerism, Reproduction in Oligochaeta.
**Arthropoda:**General characteristics and classification up to classes; Eco-Morphological notes on *Limulus, Palaemon, Cancer, Scolopendra, Julus, Bombyx, Periplaneta&Palamnaeus*
Social organization in termites and honey bee.

**UNIT- 4**

**Mollusca:** General characteristics and classification up to classes; Eco-Morphological notes on *Chiton, Dentalium, Pila, Doris, Helix, Unio, Ostrea, Pinctada, Sepia, Octopus & Nautilus.*
Torsion and Detorsion in Gastropoda, Pearl formation in Bivalves.
**Echinodermata:** General characteristics and classification up to classes; Eco-Morphological notes on *Asterias, Echinus, Cucumaria&Antedon.*
Water vascular system in Asteroidea.
Larval forms in Echinoderms.

**Instructions For Paper Setter and candidates**

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

## ZOO-DSC-1 Maj/Min-101: Diversity of Non-Chordates
## Practical

1. **Protozoa:** Study of stained slides of *Euglena, Amoeba*, *Paramecium*, *Plasmodium,* Binary fission and conjugation in *Paramecium***.**
2. **Porifera:** Study of slides of *Sycon*(T.S. and L.S.); Study of Specimens:*Sycon, Hyalonema, Euplectella&Spongilla.*
3. **Cnidaria:** Study of slides of *Obelia, Hydra;* Study of Specimens: *Physalia, Millepora, Aurelia, Tubipora, Alcyonium, Gorgonia, Metridium&Fungia.*
4. **Platyhelminthes:** Study of Specimens: *Fasciola hepatica &Taeniasolium.*
5. **Nemathelminthes:** Study of Specimens: *Ascaris&Ancylostoma*.
6. **Annelida:**Study of Specimens: *Nereis, Chaetopterus, Pheretima&Hirudinaria.*
7. **Arthropoda:**Study of Specimens: *Limulus*, *Palaemon*, *Cancer*, *Scolopendra*, *Julus*, *Bombyx, Periplaneta&Palamnaeus.*
8. **Mollusca:** Study of Specimens: *Chiton, Dentalium, Pila, Doris, Helix, Unio, Ostrea, Pinctada, Sepia, Octopus&Nautilus.*
9. **Echinodermata:** Study of Specimens: *Asterias, Echinus, Cucumaria&Antedon.*
10. **Educational trip/ Project work**.

## SUGGESTED READINGS

1. Ruppert EE and Barnes RD (2006). *Invertebrate Zoology*, VIII Edition. Holt Saunders International Edition. ISBN No.- 9780721615622.
2. Barnes RSK, Calow P, Olive PJW, Golding DW and Spicer JI (2002). *The Invertebrates: A New Synthesis*, III Edition, Blackwell Science. ISBN No.- 9781444312331.
3. Barrington EJW (1979). *Invertebrate Structure and Functions,* II Edition, Thomas Nelson & Sons Ltd. ISBN No.- 9780470265031.
4. Brusca RC, Moore W, Shuster SM (2016). *Invertebrates*, III Edition, Sinauer Associates, Inc., Publishers. ISBN No.- 9781605353753.
5. Kotpal RL (2020). *Modern Text Book of Zoology: Invertebrates,* 12th edition, Rastogi Publications. ISBN No.-9788193887554.
6. Jordan EL andVerma PS (2022). *Invertebrate Zoology,* S. Chand and Company Ltd. ISBN No.-9789355011954.
7. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Non chordates*, Pradeep Publications. ISBN No.-9789391966003.
8. Verma PS (2010). *A Manual of Practical Zoology: Invertebrates*, S. Chand and Company Ltd.ISBN No.-9788121908290.
9. Lal SS (2017). *Practical Zoology Invertebrate,* Rastogi Publications. ISBN No.-9789350781616.

## SEMESTER- II (L-100)

**Course Code:ZOO-DSC-2Maj/Min-201**
**Course Title: Diversity of Chordates**

**Total Hours= 75 (Theory= 45 + Practical= 30)** **Credits= 4 (L=3 + P=1)**
**Total Marks= 100 (Theory= 75 + Practical=25)**
**Theory Marks= 75** {67 (Theory examination) +08 (Internal assessment)}
**Practical Marks= 25** {22 (Practical examination) +03 (Internal assessment)}
**Duration for final examination: 3 hours**

### Theory

*Learning Outcomes:*

- *Students will be able to develop an appreciation for the biodiversity of chordates.*
- *Students will acquire knowledge about origin, taxonomic classification, structure and functions of chordates.*
- *Students will understand the basic characters, advancements and adaptations of different types of vertebrates.*
- *Students will be able to apply their knowledge towards identification of different species of chordates in their surroundings.*

**UNIT- 1**
**Introduction to Chordates**
General characteristics and outline classification
**Protochordata**
General characteristics of Hemichordata, Urochordata,Cephalochordata and classification upto classes; Eco-Morphological notes on*Balanoglossus, Herdmania& Amphioxus.*
Study of the larva and retrogressive metamorphosis in *Herdmania.*

**UNIT- 2**
**Agnatha**
General characteristics and classification of cyclostomes up to classes; Eco-Morphological notes on*Petromyzon&Myxine.*
**Pisces**
General characteristics of Chondrichthyes and Osteichthyes, classification up to classes; Eco-Morphological notes on*Scoliodon, Sphyrna, Pristis, Torpedo, Chimaera, Mystus, Heteropneustes, Labeo, Exocoetus, Echeneis, Anguilla, Hippocampus, Tetraodon, Diodon, Anabas&Solea*.
Migration, Osmoregulation and Parental care in fishes, Fish scales.
**Amphibia**
General characteristics and classification up to classes; Eco-Morphological notes on*Ichthyophis/Ureotyphlus, Necturus, Salamandra, Hyla, Bufo&Alytes*
Parental care in amphibians, Neoteny and Paedogenesis.

**UNIT- 3**
**Reptilia**
General characteristics and classification up to classes; Eco-Morphological notes on*Chelone, Trionyx, Hemidactylus, Varanus, Uromastix, Chamaeleon, Ophiosaurus, Draco, Ptyas, Python, Naja, Hydrophis&Crocodylus.*
Poison apparatus and biting mechanism in snakes. Identification of poisonous and non-poisonous snakes.

**Aves**

General characteristics and classification up to classes; Eco-Morphological notes on*Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis&Casuarius.*

*Archaeopteryx*- a connecting link; Principles and aerodynamics of flight, Flight adaptations and Migration in birds.

**UNIT- 4**

**Mammalia**

General characters and classification up to classes; Eco-Morphological notes on*Ornithorhynchus, Echidna*, *Macropus* (Kangaroo), *Loris, Macaca, Manis* (Scaly ant eater), *Hystrix* (porcupine), *Funambulus* (Squirrel), *Canis,Herpestes* (Mongoose), *Capra, Pteropus*.

Adaptive radiation with reference to locomotory appendages, Exoskeleton derivatives and dentition.

**Instructions For Paper Setter and candidates**

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

## ZOO-DSC-2 Maj/Min-201: Diversity of Chordates

## Practical

**1. Protochordata**
Study of specimens:*Balanoglossus, Herdmania& Amphioxus.* Larva of*Herdmania*(slide).

**2. Agnatha**
Study of specimens: *Petromyzon&Myxine.*

**3. Pisces**
Study of specimens: *Scoliodon, Sphyrna, Pristis, Torpedo, Chimaera, Mystus, Heteropneustes, Labeo, Exocoetus, Echeneis, Anguilla, Hippocampus, Tetraodon, Diodon, Anabas&Solea*.

**4. Amphibia**
Study of specimens: *Ichthyophis/Ureotyphlus, Necturus, Salamandra, Hyla, Bufo&Alytes*

**5. Reptilia**
Study of specimens: *Chelone, Trionyx, Hemidactylus, Varanus, Uromastix, Chamaeleon, Ophiosaurus, Draco, Ptyas, Python, Naja, Hydrophis&Crocodylus.*
Study of keys for the identification of poisonous and non-poisonous snakes

**6. Aves**
Study of specimens: *Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis&Casuarius.*

**7. Mammalia**
Study of specimens: *Ornithorhynchus, Echidna*, *Macropus* (Kangaroo), *Loris, Macaca, Manis* (Scaly ant eater), *Hystrix* (porcupine), *Funambulus* (Squirrel), *Canis,Herpestes* (Mongoose), *Capra &Pteropus*.

**8. Educational trip/ Project work.**

**SUGGESTED READINGS**

1. Young JZ (2004).*The Life of Vertebrates*. III Edition. Oxford University Press. ISBN No.- 9780198571735.
2. Pough H (2012).*Vertebrate life.* VIII Edition, Pearson International. ISBN No.- 9780321600790.
3. Kotpal RL (2020). *Modern Text Book of Zoology: Vertebrates, 5*th edition, Rastogi Publications. ISBN No.-9789350780954.
4. Jordan EL andVermaPS (2021). *Chordate Zoology,* S. Chand and Company Ltd. ISBN No.- 9788121916394.
5. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Chordates & Evolution*, Pradeep Publications. ISBN No.-9789391966324.
6. Verma PS (2010). *A Manual of Practical Zoology: Chordates*, S. Chand and Company Ltd.ISBN No.-9788121908306.
7. Lal SS (2015). *Practical Zoology Vertebrate,* Rastogi Publications. ISBN No.-9789350780169.

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